

Predicting Module-Lattice Reduction

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Abstract. Is module-lattice reduction better than unstructured lattice reduction? This question was highlighted as ‘Q8’ in the Kyber NIST standardization submission (Avanzi et al., 2021), as potentially affecting the concrete security of module-lattice-based schemes. Foundational works on module-lattice reduction (Lee, Pellet-Mary, Stehlé, and Wallet, ASIACRYPT 2019; Mukherjee and Stephens-Davidowitz, CRYPTO 2020) confirmed the existence of such module variants of LLL and block-reduction algorithms, but focus only on provable worst-case asymptotic behavior.

In this work, we initiate a concrete average-case analysis of module-lattice reduction. Specifically, we answer the question of the expected slope after running module-BKZ, and pinpoint the discriminant Δ_K of the number field at hand as the main quantity driving this slope. We convert this back into a gain or loss on the blocksize β : module-BKZ in a number field K of degree d requires an SVP oracle of dimension $\beta + \ln(|\Delta_K|/d^d)\beta/(d \ln \beta) + o(\beta/\ln \beta)$ to reach the same slope as unstructured BKZ with blocksize β . This asymptotic summary hides further terms that we predict concretely using experimentally verified heuristics. Incidentally, we provide the first open-source implementation of module-BKZ for (some) cyclotomic fields.

In the case of power-of-two cyclotomic conductors, we have $|\Delta_K| = d^d$, and conclude that module-BKZ needs a blocksize larger than its unstructured counterparts by $d - 1 + o(1)$. On the contrary, for all other cyclotomic fields, we have $|\Delta_K| < d^d$, so module-BKZ provides a sub-linear $\Theta(\beta/\ln \beta)$ gain on the required blocksize.

1 Introduction

Module lattices were introduced in cryptography in 1996 with the NTRU cryptosystem [HPS98], and have since seen increasing interest driven by foundational results on their average-case hardness [SSTX09, LPR10, SS11, LS15]. They are now underlying the security of three NIST post-quantum standards, ML-KEM, ML-DSA, and FN-DSA [NIS22], as well as a plethora of variants.

The central cryptanalytic tool to attack those cryptosystems is block lattice reduction, a term covering a variety of algorithms (including BKZ [Sch87, GN08b], slide [GN08a], and DBKZ [MW16]) that generalize LLL [LLL82] and offer a time-quality trade-off: roughly, block lattice reduction finds the shortest vector in a lattice in dimension n up to an approximation factor of $\Theta(\sqrt{\beta})^{n/\beta}$ in

time $2^{\Theta(\beta)}$. It operates by finding the shortest vector in *blocks* of dimension β , corresponding to projected sublattices.

Until recently, lattice-reduction attacks were mostly oblivious to the module structure of the lattice, though the potential of module variants of such algorithms was addressed in the documentation of the Kyber submission to the NIST standardization process [ABD⁺21, Sec 5.3, Q8]. Specifically, question Q8 arises from the concern that the shortest-vector subroutine could benefit from a d to d^2 speed-up factor when applied to a module lattice over a cyclotomic field of degree d [BNP17].

The discussion of Q8 [ABD⁺21, Sec 5.3, Q8] remarks that one could work over any subfield of the 512-th cyclotomic field used in Kyber, allowing to tune d to any power of 2 between 1 and 256. The choice of a large d is however highlighted as being at odds with various other speed-ups [AWHT16, LM18, Duc18]. More fundamentally, Q8 notes that the quality of such a potential module-BKZ algorithm, even using the same blocksize as BKZ, may lead to a basis of slightly worse quality when applied to module lattices over cyclotomic fields of power-of-two conductors. Given that module-lattice analogs of block lattice reduction have now been developed [LPSW19, MS20], the following question can – and should – finally be addressed:

Given the same shortest-vector oracle for lattices in dimension β , is module-BKZ better or worse than unstructured BKZ, and by how much?

This question has indeed not been fully answered in [LPSW19, MS20]: they study worst-case behavior of module-lattice analogs of LLL [LLL82] and slide reduction [GN08a], and are more concerned with the consequences of the existence of a fast short-vector oracle for module lattices of small rank over large number fields. Namely, these theoretical works present a worst-case to worst-case reduction from finding an approximately short vector in a module lattice to finding approximately short vectors in module lattices of smaller rank, up to some appropriate trade-off in the approximation factor that depends on the rank and quantities related to the number field. However, there is often a significant gap between the theoretical, worst-case understanding of block-reduction algorithms and their practical performance. Therefore, this work focuses instead on the average-case behavior of module-BKZ compared to BKZ, using a heuristic-based analysis.

1.1 Contributions

To our knowledge, this is the first study on the practical performance of module-lattice analogs of block reduction. We answer the previous question with a heuristic analysis supported by extensive experiments, providing predictions on the quality of the output basis of module-BKZ as a function of the blocksize β , measured by the *slope*. These predictions seem to fit reasonably well with our experiments, as visible in Figure 1. A small gap remains, which could be at-

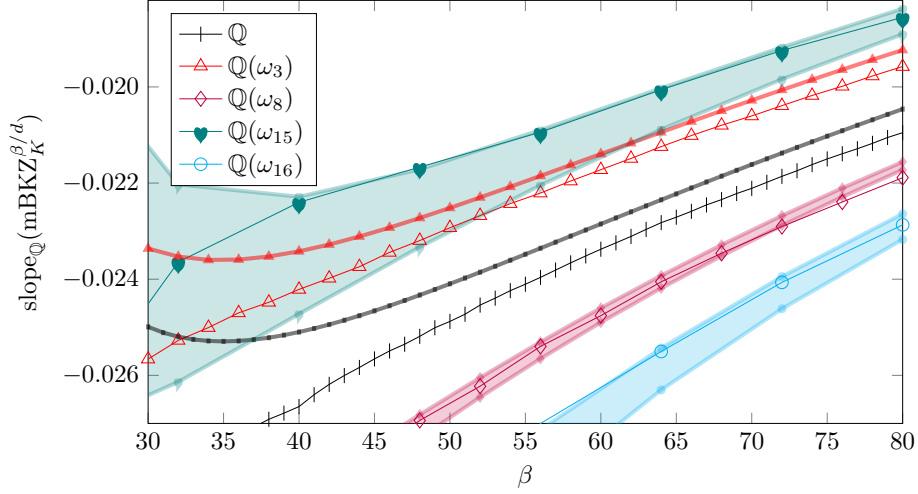


Fig. 1: Comparison of the module-BKZ $\mathbb{Q}$ -slope for several cyclotomic fields K . Measured slopes are represented by thin lines with large marks, and were averaged over 5 random lattices of $\mathbb{Q}$ -rank $r_K d = 240$ [py]. We progressively ran $5d$ tours for each multiple-of- d blocksize β to be close to convergence. Predictions [py] consist of under and over estimations, represented by thick lines with small marks and filled in between. For $\mathbb{Q}$ and $\mathbb{Q}(\omega_3)$ both predictions are too close to be distinguished.

tributed to head and tail phenomena [YD17,BSW18] that is unaccounted for by the slope model (the Geometric Series Assumption).

More precisely, we show that the ‘equivalent’ blocksize⁴ β_{eq} of module-BKZ required to match BKZ with blocksize β is asymptotically [py]:

$$\beta_{\text{eq}} = \beta + \ln \left(\frac{|\Delta_K|}{d^d} \right) \frac{\beta}{d \ln \frac{\beta}{2\pi e}} + d - 1 + o(1). \quad (1)$$

where d denotes the degree of the underlying number field K and Δ_K its discriminant. This asymptotic formula is merely meant as a summary of our analysis, as we also provide concrete predictions using explicit formulas and prediction scripts in `python`. Equation (1) shows that for a fixed number field, the gain or loss on the blocksize is barely sublinear with a constant depending on the discriminant Δ_K , and our concrete estimates show that it is quite substantial in practice, see Figure 2. It also shows that the asymptotic summary is *not* precise enough to be used as an approximation for concrete security estimates.

Incidentally, we provide an implementation of module-BKZ based on `fp111` [dt23] and `G6K` [ADH⁺19] for (some) cyclotomic fields, say up to conductor 16; it is in

⁴ For easier comparison, blocksize is measured as the $\mathbb{Q}$ -rank, not the K -rank.

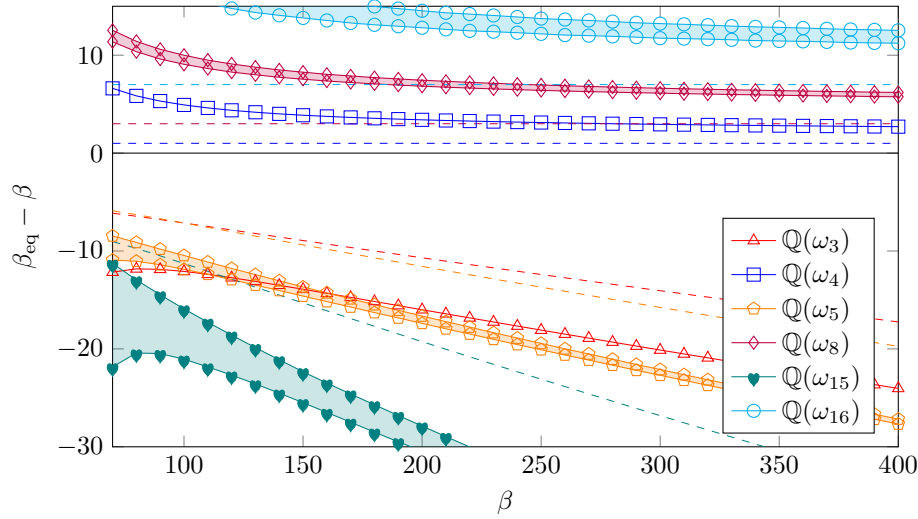


Fig. 2: Concrete predictions $[\cdot]_{py}$ for the difference of block sizes $\beta_{eq} - \beta$ for $mBKZ_K^{\beta_{eq}/d}$ to reach the same slope as unstructured BKZ^β for several cyclotomic fields K , and asymptotic predictions $[\cdot]_{py}$ in dashed lines, using formula (8) and ignoring the $o(1)$ term.

principle more general, but fails for technical limitations that we discuss later on. While our implementation is certainly far from optimized, it already allows to experiment with the behavior of module-BKZ beyond the strict focus of this paper, and should be useful for answering some of our open questions listed below. It is available at <https://github.com/lducas/mBKZ/>, and our paper systematically points to the relevant bits of our code using the symbol $[\cdot]_{py}$.

1.2 Cryptanalytic Impact

Let us now outline the impact of our work on existing cryptographic schemes. First of all, our observations are not bound to the scheme's underlying field, but can work with an arbitrary subfield of it. In particular, for a scheme over a cyclotomic field $\mathbb{Q}(\omega_s)$ of conductor s , we can work over any $\mathbb{Q}(\omega_c)$ for c dividing s . For example, for Kyber, an attacker has the freedom to work over cyclotomic fields of degree $d = 2^k$ for $k \leq 8$. A larger d should allow them to obtain speed-ups in the SVP oracle, but also restricts them to use a blocksize multiple of this larger value of d .

In the case of power-of-two cyclotomic fields, which are relevant to the new NIST standards ML-KEM, ML-DSA, and FN-DSA [NIS22], the main asymptotic term $\ln(|\Delta_K|/d^d)\beta/(d \ln \beta)$ disappears as the discriminants of these fields satisfy $|\Delta_K| = d^d$. The remaining term is $d - 1 + o(1)$, and Figure 2 shows a slow⁵

⁵ For example, for $\mathbb{Q}(\omega_{16})$, we have $d - 1 = 7$, but $\beta_{eq} - \beta \in [11.2, 12.6]$ at $\beta = 400$.

convergence from above. This confirms and quantifies the discussion of Kyber’s Q8 [ABD⁺21, Sec 5.3, Q8], which already suggested module-BKZ might need an increased blocksize compared to unstructured BKZ. Nevertheless, quite a bit of work remains to settle the question of whether module-BKZ can slightly outperform BKZ in this context, see the following Open Questions 1, 2, and 3.

On the contrary, for other cyclotomic fields, especially those whose conductor has one or more small odd prime factors, we predict a rather substantial gain on the slope, as illustrated in Figure 2. In fact, we predict that each additional odd prime factor leads to a further slope gain: for instance, compare $\mathbb{Q}(\omega_{15})$ to $\mathbb{Q}(\omega_3)$ and $\mathbb{Q}(\omega_5)$ in Figure 2. The slope gain induced by odd prime factors is relevant as popular alternatives to power-of-two conductors are conductors of the form $2^i \cdot 3^j$ [BDF18, EFG⁺22, EWY23]. In that case, module-BKZ over the third cyclotomic field is predicted to gain 20 more dimensions on the blocksize at NIST security level 1 ($\beta \approx 380$).

There are more schemes using conductors of the form $2^i \cdot 3^j$ [PFH⁺17, LS19, HWK⁺25, Kpq25]. These include an intermediate parameter set of Falcon when it was submitted to the first round of the NIST PQC standardization process, and one of the two selected Korean post-quantum PKE/KEM standards NTRU+. The Homomorphic Encryption library `HElib` also supports general conductors, and a set of parameters with a multiple-of-5 conductor was highlighted in [HS14]. However, these schemes use the Euclidean norm defined by the coefficient embedding rather than the canonical embedding, and the impact of our result would require special consideration (see Open Question 4).

Lastly, our result might narrow down avenues to construct more efficient schemes based on module-LIP [DPPvW22]. The LIP framework [DvW22] was designed to harness the decoding capabilities of dense lattices in cryptography, but competitive proposals likely require a module structure. One would naturally turn to the algebraic construction of remarkable lattices, such as the construction [BF00] of the Leech lattice as an ideal in a cyclotomic field of conductor 35, 39, 52, 56, or 84. Worse, one could be tempted to use Martinet’s construction [Mar78] of asymptotically dense lattices based on towers of number fields of bounded discriminants.

1.3 Open Questions

While the prediction and experimental analysis of the module-BKZ slope highlight its potential advantages, this study alone does not suffice to precisely quantify the cost of attacks based on module-BKZ. We list several future directions, including continued investigation of Q8 [ABD⁺21, Sec 5.3, Q8].

1. **HKZ Profile, Tails, and Dimensions for Free.** In the same way as module-BKZ affects the slope of BKZ, we also expect that the profile of module-HKZ reduction is affected by the module structure. Predicting HKZ shapes in a similar fashion would allow to replace the Geometric Series Assumption by its tail-adapted refinement [AD21]. Maybe more critically, this

change of shape should also change the number of dimensions for free in the shortest-vector subroutine [Duc18], slowing or accelerating the shortest-vector oracle in addition to affecting the BKZ slope.

2. **Simulation.** This study is limited to the slope of BKZ after convergence, i.e., after many tours. In practice, cryptanalysts are more aggressive and run only a single or a few tours, progressively increasing the blocksize [AWHT16]. The fine-tuning of attacks and security estimates then resorts to BKZ simulators [CN11, BSW18, XWW⁺24], which should be adapted to module-BKZ. In particular, one may question how fast module-BKZ converges compared to unstructured BKZ.
3. **Advanced Lattice Sieving with Cyclotomic Symmetries.** While speed-ups and memory saving of sieving has been demonstrated over cyclotomic ideal lattices [BNP17], it was done for a rather naive version of sieving [MV10]. It is far from clear whether this technique would combine well in practice with improved sieving techniques based on locality-sensitive hashing [Laa16, BDGL16] and other practical tricks used by the currently fastest sieving implementations, such as progressive sieving and the dimensions-for-free technique [LM18, Duc18, ADH⁺19, DSvW21].
4. **Coefficient Embedding.** Lastly, we note that our predicted and observed slope gains are achieved using the canonical embedding to define the geometry of cyclotomic fields. While it is algebraically more natural to use the canonical embedding, some schemes [HS14, PFH⁺17, LS19, HWK⁺25] use the coefficient embedding instead. We note that the distortion to go from one embedding to the other only depends on the number field, and is constant with growing module rank r . On the contrary, our gain on the slope leads to a gain on the first vector's length that grows exponentially with r for a fixed blocksize. Hence, if r is large enough, it will eventually be worth applying module-BKZ using the canonical embedding despite the targeted scheme using a different embedding. Nevertheless, some study is required to determine the exact break-even point and make concrete predictions for this set-up.

1.4 Technical Overview

We start by briefly recalling notions from lattice reduction: the quality of a lattice basis $(\mathbf{z}_1, \dots, \mathbf{z}_n) \subseteq \mathbb{R}^{n \times n}$ is measured by its *profile*, namely the sequence $\ln \|\mathbf{z}_1^*\|, \dots, \ln \|\mathbf{z}_n^*\|$ of logarithmic norms of its Gram-Schmidt vectors. The sum of those logarithms is an invariant of the lattice, the logarithm of its determinant. This profile is typically decreasing in i , and the flatter it is, the better. Since $\mathbf{z}_1^* = \mathbf{z}_1$, a reduced basis provides in particular a short nonzero lattice vector. Note that a vector $\mathbf{b} \in K^r$ can naturally be viewed as a vector $\mathbf{z} \in \mathbb{R}^{rd}$ through the canonical embedding, allowing us to refer to its length $\|\mathbf{b}\|$ as the Euclidean norm of $\mathbf{z}$. See Section 2 for more details.

Two Enlightening Cases. Consider an $\mathcal{O}_K$ -module lattice over the fourth cyclotomic field $K = \mathbb{Q}(\iota)$, for $\mathcal{O}_K$ its ring of integers. Having found a somewhat

short vector $\mathbf{b}_1$, we want to use it to perform module-structured lattice reduction. Naturally, one would set $\mathbf{b}_2 := \iota \mathbf{b}_1$.⁶ It is always the case that $\mathbf{b}_1$ is orthogonal to $\iota \mathbf{b}_1$, hence $\|\mathbf{b}_1^*\| = \|\mathbf{b}_2^*\|$. The profile may look perfectly flat locally, but this constraint actually makes the global profile less flat: because $\|\mathbf{b}_2^*\|$ is a bit larger than it would have been for an unstructured reduction, more of the determinant has been consumed, lowering the rest of the profile. This discussion implicitly assumes that the length of $\mathbf{b}_1$ would be the same in the structured and unstructured cases, a matter we will discuss below.

If we replace the fourth cyclotomic by the third, $\mathbb{Q}(\omega_3)$, the situation is rather different. In this case, $\mathbf{b}_1$ and $\mathbf{b}_2 := \omega_3 \mathbf{b}_1$ always form an angle of $\pi/3$, hence $\|\mathbf{b}_1^*\| = \sqrt{4/3} \cdot \|\mathbf{b}_2^*\|$, and $\mathbf{b}_2^*$ is significantly shorter than what it would be in an unstructured reduction, making the profile locally more inclined, but globally flatter.

The General Case. Consider an $\mathcal{O}_K$ -module lattice $\mathcal{M}$ for some number field K . Writing $\mathbf{b}_1$ for the first vector in a basis of $\mathcal{M}$, its length is predicted by the Gaussian Heuristic in the unstructured case, and backed by more formal theorems [Rog56, Söd11, Che13, LN20] using a careful definition of random lattices. Fortunately, an adaptation of those theorems has recently been proven for module lattices [GSVV24], which we use as a module-lattice Gaussian Heuristic in our analysis. It predicts that the shortest vector of a random cyclotomic module-lattice is only barely larger than in a random unstructured lattice.

If K is an imaginary quadratic number field, we further observe that the rank-1 module lattice $\mathbf{b}_1 \mathcal{O}_K$ is always a scaled rotation of $\mathcal{O}_K$ itself, and the quality of the first part of this basis is therefore directly related to the density of $\mathcal{O}_K$ as a lattice, or, in algebraic terms, to the absolute discriminant $|\Delta_K|$.

The situation is a bit more complex beyond imaginary quadratics, where $\mathbf{b}_1 \mathcal{O}_K$ need not be a scaled rotation of $\mathcal{O}_K$: it gets *skewed*. Yet, our heuristic analysis below (adapted from [DvW21, Lemma 4.4]) shows that as we model $\mathbf{b}_1$ as following a spherical distribution, the skewness quickly decreases as the rank increases, making this concern asymptotically irrelevant, and concretely controlled. We observe less average skewness experimentally than predicted by our model, and provide an explanation of why this model is not perfectly accurate. As we have no better model to offer, we translate this into an interval with one end corresponding to no skewness and the other corresponding to the spherical-model estimate. Luckily, the consequence of that uncertainty is quantitatively mild, fading away as the blocksize grows.

Another complication can happen, namely that $\mathbf{b}_1 \mathcal{O}_K$ does not capture all the module-lattice points in $\mathbf{b}_1 K$. What we really want to construct as the first rank-1 module is $\mathbf{b}_1 K \cap \mathcal{M}$, where $\mathcal{M}$ is the module lattice at hand. Here again, a heuristic analysis involving the Dedekind zeta function (in a fashion similar to [ABD16, DPPvW22, DvW21]) allows to model the distribution of the *index* of

⁶ Other choices $\mathbf{b}_2 := (k + \iota) \mathbf{b}_1$ for any $k \in \mathbb{Z}$ are also valid but would not change $\mathbf{b}_2^*$.

$\mathbf{b}_1\mathcal{O}_K$ in $\mathbf{b}_1K \cap \mathcal{M}$. Again, this model appears to be an overestimate compared to the index encountered experimentally, for an explainable reason, without an obvious fix, but luckily again the uncertainty it leaves is inconsequential.

This gives us four terms driving the slope of module-BKZ: the module-lattice Gaussian Heuristic, the discriminant, the skewness, and the index. Putting it all together, we conclude with a concrete slope prediction, and an asymptotic analysis of the gain or loss in terms of the blocksize.

2 Preliminaries

Notation. The set of positive (rational) integers is denoted by $\mathbb{N}$. For integers $x, x' \in \mathbb{Z}$, we define $\llbracket x; x' \rrbracket := \{x, x+1, \dots, x'\}$. We write $x \stackrel{\$}{\leftarrow} X$ to denote that x is sampled uniformly at random from X . We denote Euler’s totient function by ϕ : for $x \in \mathbb{N}$, $\phi(x)$ equals the number of integers in $\llbracket 1; x \rrbracket$ coprime to x .

Throughout the article, we use the symbol $[\text{.py}]$ for external (clickable) links to relevant parts of our implementation corresponding to the current equation or technical detail.

2.1 Lattice Background

A *lattice* in $\mathbb{R}^n$ is a set of the form $\mathcal{L} = \mathcal{L}(B) := B\mathbb{Z}^k$ for some $B \in \mathbb{R}^{n \times k}$ with linearly independent columns. We call B a *basis* for $\mathcal{L}$, and say that $\mathcal{L}$ has rank k and dimension n . If $k = n$, $\mathcal{L}$ is said to be of full rank.

An important invariant of a lattice $\mathcal{L}$ is its *first minimum*, denoted by $\lambda_1(\mathcal{L})$ and defined as $\lambda_1(\mathcal{L}) := \inf\{\|x\| : x \in \mathcal{L} \setminus \{0\}\}$, where $\|\cdot\|$ is the Euclidean norm. The task of finding a nonzero lattice vector of minimal length is known as the *Shortest Vector Problem* (SVP). In this work, we will assume that we have an SVP oracle at our disposal, i.e., an algorithm that gives us a shortest nonzero lattice vector when given a basis of a lattice.⁷

Another important invariant of a lattice $\mathcal{L}$ is its *determinant* $\det_{\mathbb{Q}}(\mathcal{L})$, also often called its *volume* (or, more accurately, its *covolume*). It is defined as $\det_{\mathbb{Q}}(\mathcal{L}) := \sqrt{|\det(B^T B)|}$ for a basis B of $\mathcal{L}$; yet its value is independent of the basis in consideration. When $\mathcal{L}$ is of full rank n , we define the *normalized* lattice $\mathcal{L}^{(1)} = \mathcal{L} / \det_{\mathbb{Q}}(\mathcal{L})^{1/n}$, so that $\mathcal{L}^{(1)}$ has determinant equal to 1.

Remark 1 (On random lattices). Our work studies the performance of module-BKZ on *random lattices*, which involves analyzing various related random lattices encountered during the algorithm. It is therefore worth mentioning what we mean with ‘random’ here. It is well known that using the Haar measure, one can formally define a uniform distribution over the set of full-rank lattices $\mathcal{L} \subseteq \mathbb{R}^n$

⁷ There exist generalizations of $\lambda_1(\mathcal{L})$ and SVP to other norms, but in this work we always consider the Euclidean distance when talking about a shortest vector.

of unit determinant [Sie45]. However, as is the case for unstructured BKZ, the distribution of the random lattices encountered in our analysis of module-BKZ is not well understood. The notion of random lattice in this work is therefore not always explicitly defined, and refers to its actual distribution induced by module-BKZ. Nevertheless, we circumvent this lack of understanding in a similar manner as is done in analysis of unstructured BKZ: we approximate these distributions using heuristics that we verify experimentally.

2.2 Algebraic Background

Let K be a number field of degree $d = [K : \mathbb{Q}]$, i.e., $K \cong \mathbb{Q}[X]/P(X)$ for some irreducible polynomial $P \in \mathbb{Q}[X]$ of degree d . K admits d distinct *embeddings* into $\mathbb{C}$, i.e., injective field homomorphisms from K to $\mathbb{C}$. Each of these embeddings corresponds to evaluating elements of K at one of the roots of P in $\mathbb{C}$. We have $d = d_{\mathbb{R}} + 2d_{\mathbb{C}}$, where $d_{\mathbb{R}}$ denotes the number of real embeddings (corresponding to the roots of P in $\mathbb{R}$) and $d_{\mathbb{C}}$ the number of complex embeddings, up to conjugation. We denote the set of real embeddings by $\mathcal{E}_{\mathbb{R}}$, the $d_{\mathbb{C}}$ embeddings corresponding to roots with strictly positive imaginary part by $\mathcal{E}_{\mathbb{C}}^+$, and their conjugates by $\overline{\mathcal{E}_{\mathbb{C}}^+}$. Altogether, the set $\mathcal{E}$ of all d embeddings of K decomposes as $\mathcal{E} = \mathcal{E}_{\mathbb{R}} \uplus \mathcal{E}_{\mathbb{C}}^+ \uplus \overline{\mathcal{E}_{\mathbb{C}}^+}$.

We define the ring $K_{\mathbb{R}}$ as $K \otimes_{\mathbb{Q}} \mathbb{R}$. Note that we have a ring isomorphism $K_{\mathbb{R}} \cong \mathbb{R}^{d_{\mathbb{R}}} \times \mathbb{C}^{d_{\mathbb{C}}}$. We define the field *trace* by $\text{Tr}: K_{\mathbb{R}} \rightarrow \mathbb{R}, x \mapsto \sum_{j=1}^d \sigma_j(x)$, inducing an inner product on $K_{\mathbb{R}}$ given by $\langle x, y \rangle := \text{Tr}(x\bar{y})$ for $x, y \in K_{\mathbb{R}}$. This trace inner product is extended to $K_{\mathbb{R}}^m$ by defining $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{i=1}^m \langle x_i, y_i \rangle = \sum_{i=1}^m \text{Tr}(x_i \bar{y}_i)$ for $\mathbf{x} = (x_1, \dots, x_m), \mathbf{y} = (y_1, \dots, y_m) \in K_{\mathbb{R}}^m$. (In particular, this yields a geometric norm $\|\cdot\|: \mathbf{x} \mapsto \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$.) We also define $\langle \mathbf{x}, \mathbf{y} \rangle_K := \sum_{i=1}^m x_i \bar{y}_i \in K_{\mathbb{R}}$.

Besides the trace norm $\|x\| = \sqrt{\langle x, x \rangle}$ of an element $x \in K_{\mathbb{R}}$, we consider the *algebraic norm* $N(x) := \prod_{i=1}^d \sigma_i(x)$. By the arithmetic-geometric inequality, we have $\sqrt{d}N(x)^{1/d} \leq \|x\|$. For $\mathbf{x} \in K_{\mathbb{R}}^m$, we write $N(\mathbf{x})$ as shorthand for $N(\langle \mathbf{x}, \mathbf{x} \rangle_K)^{1/2}$.

We write $\mathcal{O}_K$ for the *ring of integers* of K , which consists of the elements $x \in K$ such that $Q(x) = 0$ for some monic polynomial $Q \in \mathbb{Z}[X]$. In particular, $\mathcal{O}_K$ is a free $\mathbb{Z}$ -module of rank d , i.e., it is the set of all $\mathbb{Z}$ -linear combinations of a basis $(o_1, \dots, o_d)$ in $\mathcal{O}_K$ (called a $\mathbb{Z}$ -basis or integral basis for $\mathcal{O}_K$); for instance, $(1, X, \dots, X^{d-1})$ is such a basis. The ring of integers $\mathcal{O}_K$ together with the embeddings $\sigma_1, \dots, \sigma_d$ allow us to define the *discriminant* Δ_K of the number field K , which is given by $\Delta_K := |\det((\sigma_i(o_j))_{i,j})|^2$.

We denote by μ_K the number of roots of unity in K .

Cyclotomic Fields. The main class of number fields we consider in this work is the class of *cyclotomic fields* [py], which are number fields of the form $K = \mathbb{Q}(\omega_c)$ for $c \in \mathbb{N}$, where ω_c is a primitive c -th root of unity (i.e., $\omega_c \in \mathbb{C}$ and satisfies $\omega_c^c = 1$ and $\omega_c^j \neq 1$ for all $1 \leq j < c$). The *conductor* of K is the minimal $c \in \mathbb{N}$

such that $K = \mathbb{Q}(\omega_c)$. If $c \in \mathbb{N}$ is odd, then $\mathbb{Q}(\omega_c) \cong \mathbb{Q}(\omega_{2c})$, and this criterion captures all isomorphic cyclotomic fields: the conductor c of a cyclotomic field K is either odd (in which case $K \cong \mathbb{Q}(\omega_{2c})$) or a multiple of 4. The number of roots of unity in a cyclotomic field K of conductor c equals $\mu_K = c$ when c is even, and $\mu_K = 2c$ when c is odd.

A cyclotomic field $K = \mathbb{Q}(\omega_c)$ satisfies $K \cong \mathbb{Q}[X]/\Phi_c(X)$ for the c -th cyclotomic polynomial Φ_c . The degree of $K = \mathbb{Q}(\omega_c)$ equals $\phi(c)$, and its ring of integers is $\mathbb{Z}[\omega_c]$, which is isomorphic to $\mathbb{Z}[X]/\Phi_c(X)$. When $c > 2$, all of K 's embeddings are complex: $d_{\mathbb{R}} = 0$ and $d_{\mathbb{C}} = d/2$.

Ideals, Modules, and Module Lattices. A *fractional ideal* of $\mathcal{O}_K$ is an $\mathcal{O}_K$ -submodule $\mathfrak{I} \subseteq K$ (i.e., it is closed under addition and under multiplication by elements of $\mathcal{O}_K$) for which there exists $x \in K \setminus \{0\}$ satisfying $x\mathfrak{I} \subseteq \mathcal{O}_K$. The algebraic norm $N(\mathfrak{I})$ of a fractional ideal $\mathfrak{I}$ is defined as $N(\mathfrak{I}) := [\mathcal{O}_K : x\mathfrak{I}] / |N(x)|$ for any $x \in K \setminus \{0\}$ satisfying $x\mathfrak{I} \subseteq \mathcal{O}_K$.

An $\mathcal{O}_K$ -*module* is a set of the form $\mathcal{M} = \sum_{i=1}^r \mathbf{b}_i \mathfrak{I}_i$ for nonzero fractional $\mathcal{O}_K$ -ideals $\mathfrak{I}_1, \dots, \mathfrak{I}_r$ and $K_{\mathbb{R}}$ -linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_r \in K_{\mathbb{R}}^m$ (for some $m > 0$). We say that $\mathcal{M}$ is a *free* module if the $\mathfrak{I}_i$'s are all equal to $\mathcal{O}_K$. We refer to the set $(\mathbf{b}_i, \mathfrak{I}_i)_{i=1}^r$ as a *pseudobasis* for $\mathcal{M}$, and remark that another way to represent module lattices is using module filtrations (see [MS20]). Such a pseudobasis is said to be *unital* if $1 \in \mathfrak{I}_i$ for all i (equivalently, if $\mathcal{O}_K \subseteq \mathfrak{I}_i$ for all i).⁸ We remark that $\mathcal{O}_K \subseteq \mathfrak{I}_i$ implies that $N(I_i) \leq 1$.

The rank of $\mathcal{M}$ is defined to be $\text{rank}_{K_{\mathbb{R}}}(\mathcal{M}) := \dim_{K_{\mathbb{R}}} \text{span}_{K_{\mathbb{R}}}(\mathcal{M})$. For simplicity, we take $m = r = \text{rank}_{K_{\mathbb{R}}}(\mathcal{M})$ in the remainder of this work. Denoting by $\mathbf{B}$ the matrix with columns $\mathbf{b}_1, \dots, \mathbf{b}_r$, the determinant of $\mathcal{M}$ in $K_{\mathbb{R}}$ is defined as $\det_{K_{\mathbb{R}}}(\mathcal{M}) = \det(\overline{\mathbf{B}}^T \mathbf{B})^{1/2} \prod_{i=1}^r \mathfrak{I}_i$.

Through the embeddings of K , one can view $\mathcal{M}$ as an rd -dimensional lattice in $\mathbb{C}^{rd}$.⁹ Specifically, writing $\boldsymbol{\sigma}: K \rightarrow \mathbb{C}^d$ for the *canonical embedding*, i.e., $\boldsymbol{\sigma}(x) = (\sigma(x))_{\sigma \in \mathcal{E}}$, the corresponding module lattice consists of vectors of the form $(\boldsymbol{\sigma}(x_1), \dots, \boldsymbol{\sigma}(x_r)) \in \mathbb{C}^{rd}$ for vectors $(x_1, \dots, x_r) \in \mathcal{M} \subseteq K^r$. (For instance, taking $K = \mathbb{Q}$ and $R = \mathbb{Z}$, the canonical embedding is trivial and we recover the usual notion of Euclidean lattices in $\mathbb{Q}^r$.) Note that this embedding preserves the geometry: the trace inner product of the module vectors in K^r is exactly the Euclidean inner product of the corresponding vectors in $\mathbb{C}^{rd}$. With abuse of notation, we write $\mathcal{M}$ for both the module and the module lattice corresponding to it. It can be shown that the determinant of the module lattice $\mathcal{M}$ equals $\det_{\mathbb{Q}}(\mathcal{M}) := |\Delta_K|^{r/2} N(\det_{K_{\mathbb{R}}}(\mathcal{M}))$.¹⁰

⁸ The scaling algorithm from [LPSW19, Alg. 3.2] turns an arbitrary pseudobasis into a unital pseudobasis.

⁹ Since $\mathbb{C}^{rd} \cong \mathbb{R}^{2rd}$, there is a natural relation with the lattice definition in Section 2.1.

¹⁰ Sometimes a different normalization of embeddings is used (e.g., [PM19]), resulting in an additional factor of $2^{-rd_{\mathbb{C}}}$.

2.3 GSO over $K_{\mathbb{R}}$

Two vectors are said to be $K_{\mathbb{R}}$ -linearly independent if and only if the zero vector cannot be expressed as a non-trivial $K_{\mathbb{R}}$ -linear combination of them. This notion allows us to extend the Gram-Schmidt orthogonalization procedure to matrices $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_r) \in K_{\mathbb{R}}^{m \times r}$ with $K_{\mathbb{R}}$ -linearly independent columns $\mathbf{b}_i$. (For instance, see [FS10, LPSW19].)

Definition 2 (GSO over $K_{\mathbb{R}}$). Given $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_r)$ such that the $\mathbf{b}_i$ are $K_{\mathbb{R}}$ -linearly independent, we define its GSO as $\mathbf{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_r^*)$, where $\mathbf{b}_1^* = \mathbf{b}_1$ and, for all $1 < i \leq r$,

$$\mathbf{b}_i^* := \mathbf{b}_i - \sum_{j=1}^{i-1} \mu_{i,j} \mathbf{b}_j^* \quad \text{with } \mu_{i,j} = \frac{\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle_K}{\langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle_K} \text{ for all } 1 \leq j < i.$$

More generally, we define the projection $\pi_i: \mathbf{x} \mapsto \mathbf{x} - \sum_{j=1}^{i-1} \frac{\langle \mathbf{x}, \mathbf{b}_j^* \rangle_K}{\langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle_K} \mathbf{b}_j^*$ for all i , so $\mathbf{b}_i^* = \pi_i(\mathbf{b}_i)$. Note that $\langle \mathbf{b}_i^*, \mathbf{b}_j^* \rangle_K = 0$ for all $i \neq j$.

Next, let $\mathcal{M}$ be a rank- r module lattice with pseudobasis $\mathfrak{B} = ((\mathbf{b}_i, \mathfrak{J}_i))_{i=1}^r$. For $1 \leq i \leq j \leq r$, we define the *projected module lattice* $\mathcal{M}(\mathfrak{B}_{[i;j]}) = \pi_i(\mathbf{b}_i)\mathfrak{J}_i + \dots + \pi_i(\mathbf{b}_j)\mathfrak{J}_j$. Note that $\mathcal{M}(\mathfrak{B}_{[i;j]})$ has rank $j - i + 1$, and depends on the pseudobasis at hand.

The determinants of $\mathcal{M}$ and its projected module lattices can be expressed in terms of the GSO of the pseudobasis. Namely, for all $1 \leq i \leq j \leq r$, we have:

$$\det_{\mathbb{Q}}(\mathcal{M}(\mathfrak{B}_{[i;j]})) = |\Delta_K|^{(j-i+1)/2} \cdot \prod_{k=i}^j N(\mathbf{b}_k^*) N(\mathfrak{J}_k).$$

3 Module-BKZ

We first review BKZ for arbitrary lattices and its corresponding slope prediction, as our analysis of the modified BKZ algorithm for module lattices, *module-BKZ*, follows a similar pattern. Next, we describe module-BKZ, and proceed with the specifics of our implementation.

3.1 BKZ and Corresponding Slope Prediction

Given a basis of an n -dimensional lattice with Gram-Schmidt orthogonalization (GSO) $(\mathbf{b}_1^*, \dots, \mathbf{b}_n^*)$, we define its $\mathbb{Q}$ -*profile*¹¹ as the sequence $(\ell_1^{\mathbb{Q}}, \dots, \ell_n^{\mathbb{Q}})$, where $\ell_i^{\mathbb{Q}} := \ln \|\mathbf{b}_i^*\|$. As aforementioned, the more balanced the profile is, the better.

¹¹ This is usually simply called the (log-)profile, but as we will generalize this notion to other number fields we explicit the number field for unambiguous discussions.

Turning a basis into a basis of better quality is the task of lattice reduction algorithms, such as the BKZ algorithm [Sch87, SE94]. In short, for blocksize β , the BKZ algorithm aims to return a lattice basis $\mathbf{B}$ such that its GSO $(\mathbf{b}_1^*, \dots, \mathbf{b}_n^*)$ is BKZ^β -reduced, i.e., it satisfies

$$\|\mathbf{b}_i^*\| = \lambda_1(\mathcal{L}(\mathbf{B}_{\llbracket i; \min(i+\beta-1, n) \rrbracket})) \quad \forall 1 \leq i \leq n,$$

where $\mathcal{L}(\mathbf{B}_{\llbracket i; j \rrbracket})$ is the lattice generated by $(\pi_i(\mathbf{b}_i), \dots, \pi_i(\mathbf{b}_j))$ and π_i is the linear projection orthogonal to the span of $\mathbf{b}_1, \dots, \mathbf{b}_{i-1}$. (Approximation variants of BKZ exist as well, but are beyond the scope of this paper.)

Slope Prediction under the Geometric Series Assumption. One can predict the slope of the $\mathbb{Q}$ -profile of a BKZ^β -reduced basis. These predictions are often based on the empirical observation that the $\mathbb{Q}$ -profile of a BKZ^β -reduced basis tends to resemble a descending straight line for sufficiently large $\beta \ll n$ (say $\beta \geq 50$). This is formalized by the Geometric Series Assumption (GSA), as first proposed by Schnorr [Sch03].

Heuristic 1 (GSA for Euclidean Lattices) *Let $\beta \ll n$ be sufficiently large. There is a constant $\alpha_{\mathbb{Q}} > 1$ (depending only on β) such that the BKZ^β -reduced basis of a random full-rank lattice $\mathcal{L} \subseteq \mathbb{R}^n$ satisfies:*

$$\mathbb{E}[\ell_i^{\mathbb{Q}}] = \mathbb{E}[\ell_1^{\mathbb{Q}}] - (i-1) \ln \alpha_{\mathbb{Q}} \quad \forall 1 \leq i \leq n. \quad (\text{GSA})$$

In fact, due to the lattice invariant $\ln \det_{\mathbb{Q}}(\mathcal{L}) = \sum_{i=1}^n \ell_i^{\mathbb{Q}}$ and the definition of BKZ reduction, the GSA is equivalent to the following prediction of the *expected $\mathbb{Q}$ -slope*, defined as $\text{slope}_{\mathbb{Q}}(\text{BKZ}^\beta) := -\ln \alpha_{\mathbb{Q}}$. (See, for example, [AD21, dBvW25].)

Heuristic Claim 1 ($\mathbb{Q}$ -profile of BKZ^β under GSA) *Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a random full-rank lattice of fixed determinant, and let $\mathbf{B}$ be a BKZ^β -reduced basis of $\mathcal{L}$ for some sufficiently large $\beta \ll n$. Then the GSA predicts*

$$\mathbb{E}[\ell_i^{\mathbb{Q}}] = \frac{n+1-2i}{2} \ln \alpha_{\mathbb{Q}} + \frac{1}{n} \ln \det_{\mathbb{Q}}(\mathcal{L})$$

for all $1 \leq i \leq n$, where:

$$\ln \alpha_{\mathbb{Q}} = \frac{2}{\beta-1} \mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]$$

where $\mathbf{s}$ is a shortest vector in one of the random normalized projected lattices $\mathcal{L}(\mathbf{B}_{\llbracket j; j+\beta-1 \rrbracket})^{(1)}$ for $1 \leq j < n - \beta$.

Further Prediction using the Gaussian Heuristic. In the analysis of BKZ algorithms, it is standard to go one step further in the slope prediction by assuming that the normalized projected lattices $\mathcal{L}(\mathbf{B}_{\llbracket j; j+\beta-1 \rrbracket})^{(1)}$ behave as a random

rank- β lattice of unit determinant, allowing to invoke the Gaussian Heuristic to estimate $\mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]$.

More precisely, the Gaussian Heuristic states that, for a rank- n lattice $\mathcal{L}$ and a measurable set $\mathcal{B} \subseteq \text{span}(\mathcal{L})$, the number of nonzero lattice vectors in $\mathcal{L} \cap \mathcal{B}$ approximately equals $\text{vol}(\mathcal{B})/\det_{\mathbb{Q}}(\mathcal{L})$. Applying this heuristic to the n -dimensional Euclidean unit ball $\mathcal{B}_n$ gives a prediction of $\lambda_1(\mathcal{L}) \approx (\det_{\mathbb{Q}}(\mathcal{L})/\text{vol}(\mathcal{B}_n))^{-1/n}$ for the first minimum of $\mathcal{L}$.

For carefully defined notions of random lattices, there exists a more precise and formal result giving the expectation of λ_1 ; for example, see [BSW18, Theorem 1] (attributed to [Söd11, Che13]). Note that this formal result still downgrades to the following *heuristic* when applied to the lattices appearing in BKZ, whose distributions are not yet well understood.

Heuristic 2 ($\lambda_1(\mathcal{L})$ under the Gaussian Heuristic for $\mathcal{L} \subseteq \mathbb{R}^n$ [py]) *Let $\mathcal{L}$ be a random full-rank lattice of unit determinant in $\mathbb{R}^n$. Its expected first minimum under the Gaussian Heuristic is given by*

$$\mathbb{E}[\lambda_1(\mathcal{L})] = \text{gh}(n) \tag{GH}$$

where $\text{gh}(n) := 2^{1/n} \cdot \Gamma(1 + 1/n) \cdot \text{vol}(\mathcal{B}_n)^{-1/n}$.

Remark 3 (Gaussian Heuristic for Lattices of Arbitrary Determinants). Heuristic 2 is given for lattices of unit determinant, but applies to lattices of arbitrary determinant after appropriate scaling: given a random lattice $\mathcal{L}$ drawn from a distribution over full-rank lattices in $\mathbb{R}^n$ with fixed determinant D , the Gaussian Heuristic predicts: $\mathbb{E}[\lambda_1(\mathcal{L})] = \text{gh}(n) \cdot D^{1/n}$.

Note that $\text{gh}(n) \approx \sqrt{n/(2\pi e)}$. Defining $\text{lgh}_{\mathbb{Q}}(n) := \ln \text{gh}(n)$, we can then heuristically predict the slope of the $\mathbb{Q}$ -profile as

$$\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta}) = -\frac{2}{\beta - 1} \text{lgh}_{\mathbb{Q}}(\beta).$$

3.2 Module-BKZ

Applying the usual BKZ algorithm to a rank- r module $\mathcal{M} \subseteq K_{\mathbb{R}}^r$ merely views the module $\mathcal{M}$ as an rd -dimensional lattice where $d = \deg(K)$, thereby ignoring the underlying module structure. Instead, we consider a generalization of the BKZ algorithm, which we call $\text{mBKZ}_K^{\beta_K}$ for a parameter $2 \leq \beta_K \leq r$ and present in Algorithm 1. This module-BKZ algorithm finds short vectors in the rank- r module using an SVP oracle that finds short vectors in rank- β_K projected modules, thereby generalizing module-LLL [LPSW19] to $\beta_K \geq 2$. The projected modules correspond to $\beta_K d$ -dimensional lattices, and thus $\text{mBKZ}_K^{\beta_K}$ is naturally compared to $\text{BKZ}^{\beta_K d}$.

Specifically, $\text{mBKZ}_K^{\beta_K}$ turns a pseudobasis of $\mathcal{M}$ into an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis, defined as follows.

Definition 4 (mBKZ $_{\beta_K}^{\beta_K}$ -reduced). Let $2 \leq \beta_K \leq r$. We say that a pseudobasis $\mathfrak{B}$ of a rank- r module $\mathcal{M}$ is mBKZ $_{\beta_K}^{\beta_K}$ -reduced if

$$\|\mathbf{b}_i^*\| = \lambda_1(\mathcal{M}(\mathfrak{B}_{\llbracket i; \min(i+\beta_K-1, r) \rrbracket})) \quad \forall 1 \leq i \leq r.$$

Algorithm 1: Module-BKZ algorithm (high-level)

Input: Unital pseudobasis $\mathfrak{B}$ of a rank- r module $\mathcal{M}$; parameter β_K

Output: Unital mBKZ $_{\beta_K}^{\beta_K}$ -reduced pseudobasis

```

1 while  $\mathfrak{B}$  is not mBKZ $_{\beta_K}^{\beta_K}$ -reduced do
2   for  $i = 1, \dots, r$  do
3     Let  $\mathbf{v} = \text{SVP}(\mathcal{M}(\mathfrak{B}_{\llbracket i; \min(i+\beta_K-1, r) \rrbracket}))$ 
4     Let  $\mathfrak{J}$  be such that  $\mathbf{v}\mathfrak{J} = \mathbf{v}K \cap \mathcal{M}(\mathfrak{B}_{\llbracket i; \min(i+\beta_K-1, r) \rrbracket})$ 
5     Lift  $\mathbf{v}\mathfrak{J}$  to  $\mathbf{w}\mathfrak{J} \subseteq \mathcal{M}$  satisfying  $\pi_i(\mathbf{w}) = \mathbf{v}$ 
6     Insert  $(\mathbf{w}, \mathfrak{J})$  into  $\mathfrak{B}$  at position  $i$  and remove linear dependencies
7 return  $\mathfrak{B}$ 

```

We refer to a single execution of the while-loop as an *mBKZ tour*. Note that Line 5 is essentially Babai lifting [Bab86], and Line 6 can be achieved using (e.g.) the algorithm from [FS10, Theorem 4]. While Algorithm 1 may encounter non-unital pseudobases during an mBKZ tour, the final output is always unital by construction. In contrast to formal variants of LLL and BKZ, which include a parameter $\delta < 1$ to enforce significant improvement at each insertion step, we are not concerned with guaranteed termination. Instead, we take $\delta = 1$ and simply assume approximate convergence after sufficiently many mBKZ tours. This approach is at least heuristically supported by [HPS11].

Remark 5. Note that the fractional ideal $\mathfrak{J}$ considered in Line 4 contains $\mathcal{O}_K$ and is the ideal of minimal norm that satisfies $\mathbf{v}\mathfrak{J} \subseteq \mathcal{M}(\mathfrak{B}_{\llbracket i; \min(i+\beta_K-1, n) \rrbracket})$. Indeed, consider a vector $\mathbf{v} \in \mathcal{M}'$ for some module $\mathcal{M}'$, and let $\mathfrak{J}$ be the fractional ideal such that $\mathbf{v}\mathfrak{J} = \mathbf{v}K \cap \mathcal{M}'$. Then $\mathfrak{J} = \{x \in K : x\mathbf{v} \in \mathcal{M}'\}$. Since $x\mathbf{v} \in \mathcal{M}'$ for all $x \in \mathcal{O}_K$, we have that $\mathcal{O}_K \subseteq \mathfrak{J}$, so $\mathcal{N}(\mathfrak{J}) \leq 1$. Moreover, for all fractional ideals $\mathfrak{J}'$ such that $\mathbf{v}\mathfrak{J}' \subseteq \mathcal{M}'$, we have $\mathfrak{J}' \subseteq \mathfrak{J}$, and thus $\mathcal{N}(\mathfrak{J}) \leq \mathcal{N}(\mathfrak{J}')$.

3.3 Implementation

For our experiments, we developed a **python** implementation of module-BKZ for cyclotomic fields, which is publicly available, together with the experiment and prediction scripts, and the generated data. Instead of redeveloping the entire lattice-reduction stack to obtain one specialized to module lattices, we rely on existing libraries **fp111/fpy111** [dt23] and **G6K** [ADH⁺19]. When implementing the SVP oracle from G6K_[.py], we choose less aggressive strategies than typically done to be almost certain to really solve SVP, as our purpose is more to understand and predict than to fine-tune for speed.

The module lattices considered in our experiments are natural $\mathcal{O}_K$ -analogs of q -ary lattices, namely lattices defined by random linear equations modulo $q\mathcal{O}_K$ [py], though we remark that our implementation is not restricted to those.

The choice of using existing lattice-reduction libraries raises two technical difficulties: protecting the module structure from the whims of libraries that are designed to only consider $\mathbb{Z}$ -bases, and having a geometrically faithful embedding of the cyclotomic rings that remains integral. We discuss our solutions to both issues.

Restructured $\mathbb{Z}$ -bases. Our objective is to represent a module lattice $\mathcal{M}$ by a $\mathbb{Z}$ -basis, while maintaining enough of its structure. Naively, one would be tempted to enforce that, for all blocks i , $(\mathbf{b}_{1+id}, \dots, \mathbf{b}_{d+id})$ $\mathbb{Z}$ -generates a rank-1 module, but this constraint may be broken by size reduction, the most frequent basis maintenance task of the lattice-reduction library. Size reduction can hardly be avoided for reasons of numerical stability, and replacing it by a module-lattice-analog of size reduction would require significant modifications in `fp111`.

Instead, we want the property that $(\mathbf{b}_1, \dots, \mathbf{b}_d)$ is indeed the $\mathbb{Z}$ -basis of a rank-1 module [py], and that this property holds recursively for the rest of the basis (i.e., $\mathbf{b}_{d+1}, \dots, \mathbf{b}_{rd}$) projected orthogonally to $\mathbf{b}_1, \dots, \mathbf{b}_d$. This property is not affected by size reduction, and is sufficient to implement module-BKZ.

This structure is going to be broken whenever we solve SVP on a block by `G6K`. We now want a procedure to repair the module structure; more precisely, given a certain first vector $\mathbf{b}_1$, we would like to enforce that $(\mathbf{b}_1, \dots, \mathbf{b}_d)$ is a $\mathbb{Z}$ -basis of $\mathbf{b}_1 K \cap \mathcal{M}$ [py]. This is a rank-1 module containing $\mathbf{b}_1 \mathcal{O}_K$. So we first insert a $\mathbb{Z}$ -basis of $\mathbf{b}_1 \mathcal{O}_K$, and would like to run LLL to eliminate the linear dependencies [py].

Yet, LLL may find shorter vectors for this block and therefore re-break the module structure of the first block. In principle, one can prevent LLL from doing just that while still doing the linear dependency elimination, simply by running LLL with a δ parameter very close to 0.

Unfortunately, the `fp111` API forbids setting $\delta \leq 1/4$; the reason might be that there is no absolute guarantee in term of basis quality for such small δ . Yet, for such parameters LLL still guarantees that the potential does not increase and that linear dependencies are eliminated. We argue that it would be desirable to relax the API.

Fortunately, we do not have to resort to patching `fp111` to get our prototype running, at least for some cyclotomic rings of interest (say with conductor ≤ 16); we found out experimentally that repeated attempts to restructure, with decreasing δ from 0.99 would eventually succeed before reaching $1/4$ [py]. Running BKZ progressively with increasing blocksize also seems to help avoiding the issue.

Cyclic Embedding. A second technical difficulty with our approach comes from the definition of inner products. We are working in the cyclotomic ring $\mathbb{Z}[\omega_c]$ with the trace inner product. However, the `fp111` library requires the input lattice

to be represented with integer coefficients, and considers the standard inner product over $\mathbb{R}^n$. If we represent $\mathbb{Z}[\omega_c] \cong \mathbb{Z}[X]/\Phi_c(X)$ using the coefficients of a polynomial $\sum_{i=0}^{c-1} a_i X^i$, then the inner product does not correspond to the desired trace inner product. We could represent a vector by its embeddings to have the correct inner product, but those values are defined over $\mathbb{C}$ and thus may be irrational. The following *cyclic embedding* solves this conundrum; an idea that already underpins several existing works [DD12, BDF18, DP16].

Consider the cyclic ring $\mathcal{C}_c = \mathbb{Q}[X]/(X^c - 1)$, and view it as a Euclidean vector space in the obvious way. That is, elements of $\mathcal{C}_c$ are polynomials $\sum_{i=0}^{c-1} a_i X^i$ represented by their coefficients a_i , naturally leading to the *coefficient inner product* $\langle \sum_{i=0}^{c-1} a_i X^i, \sum_{i=0}^{c-1} b_i X^i \rangle_{\text{coef}} := \sum_{i=0}^{c-1} a_i b_i$. The discrete Fourier transform F_c sends such a polynomial $P(X) = \sum_{i=0}^{c-1} a_i X^i$ (more precisely, its coefficients) to the vector $(P(\omega_c^j))_{j=0}^{c-1} \in \mathbb{C}^c$. It is well known that F_c is a scaled isometry; more precisely, its scaling $\frac{1}{\sqrt{c}} F_c$ is an isometry.

Similarly, the trace inner product of $\mathbb{Q}(\omega_c)$ is isometric to the standard inner product of $\mathbb{C}^{\phi(n)}$ via the canonical embedding. The latter Hermitian space is trivially embedded into $\mathbb{C}^c$ by padding $c - \phi(c)$ many zero coordinates at positions corresponding to the non-primitive c -th roots of unity. This allows to define the cyclic embedding by completing a commutative diagram, as depicted in Figure 3 [FJ].

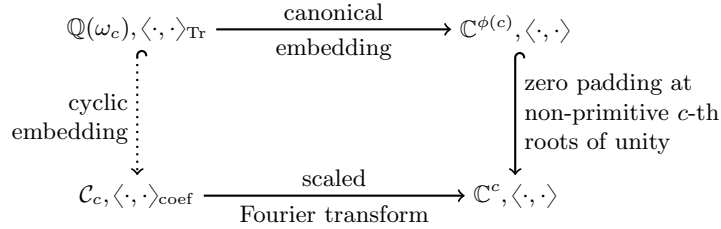


Fig. 3: Commutative diagram defining the cyclic embedding.

Being a bit more explicit in those calculations, one further notices that the integers $\mathbb{Z}[\omega_c]$ are represented by elements in $\frac{1}{c}\mathbb{Z}[X]/(X^c - 1)$; following [DvW18], this is easy to show for prime conductor c and the general case follows by direct-sum and tensor structures. Hence, scaling the representation by a factor c results in a scaled isometric embedding of $\mathbb{Z}[\omega_c]$ in $\mathbb{Z}^c$. Note that the embedding dimension c is strictly larger than the degree $\phi(c)$ of the number field, but this is not an issue as the lattice-reduction library is perfectly capable of working with non-full-rank lattices.

4 Prediction of mBKZ Profile

In this section, we present a slope prediction for (structured) module-BKZ, motivated by numerical observations and supported by theoretical analysis, which makes use of a module-lattice analog of the Geometric Series Assumption (GSA).

Let K be a number field of degree d and discriminant Δ_K . Consider a random rank- r module lattice $\mathcal{M}$, and a sufficiently large β_K satisfying $2 \leq \beta_K \ll r$. Viewing $\mathcal{M}$ as an rd -dimensional lattice $\mathcal{L}$, the analysis from Section 3.1 (using the GSA) implies that the unstructured $\text{BKZ}^{\beta_K d}$ algorithm results in a slope prediction of

$$\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta_K d}) = -\frac{2}{\beta_K d - 1} \mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|] \quad (2)$$

where the random variable $\mathbf{s} \in \mathbb{R}^{\beta_K d}$ reflects the behavior of BKZ. Specifically, $\mathbf{s}$ is a shortest vector in a random rank- $\beta_K d$ lattice of unit determinant. We recall from Section 3.1 that by invoking the Gaussian Heuristic one can predict $\mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]$ as $\text{lgh}_{\mathbb{Q}}(\beta_K d)$, as is common in BKZ-slope analysis.

In Section 4.1, we use a module-lattice analog of the GSA to derive the following slope prediction $[\text{py}]$ for the $\text{mBKZ}_K^{\beta_K}$ algorithm:

$$\begin{aligned} \text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = & -\frac{2}{\beta_K d - d} \left(\underbrace{\mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]}_{t_1} + \underbrace{\frac{1}{2d} \ln \frac{|\Delta_K|}{d^d}}_{t_2} \right. \\ & \left. + \underbrace{\mathbb{E}_{\mathbf{s}} \left[\ln \frac{\sqrt{d} N(\mathbf{s})^{1/d}}{\|\mathbf{s}\|} \right]}_{t_3} + \underbrace{\frac{1}{d} \mathbb{E}_{\mathfrak{J}}[\ln N(\mathfrak{J})]}_{t_4} \right) \quad (3) \end{aligned}$$

where the random variables $\mathbf{s} \in K_{\mathbb{R}}^{\beta_K}$ and $\mathfrak{J} \supseteq \mathcal{O}_K$ reflect the behavior of module-BKZ. Specifically, $\mathbf{s}$ is a shortest vector in a random rank- β_K module lattice of unit determinant, and $\mathfrak{J}$ is a fractional ideal in the random mBKZ-reduced pseudobasis.¹²

Apart from the terms t_1, t_2, t_3, t_4 , the slope predictions of unstructured BKZ in Equation (2) and structured module-BKZ in Equation (3) also differ in the denominator, changing from $\beta_K d - 1$ to $\beta_K d - d$.

After deriving this slope prediction, we dive into the four main terms t_1, t_2, t_3, t_4 , enabling us to compare with the usual BKZ slope prediction from Equation (2). We refer to these four terms as:

t_1 : the module-lattice analog of $\text{lgh}_{\mathbb{Q}}(\beta_K d)$,

¹² As is standard in the analysis of BKZ algorithms, these distributions are not formally defined, which is why most of our analysis remains heuristic (recall Remark 1).

t_2 : the discriminant gap,

t_3 : the skewness gap,

t_4 : the index gap.

Note that t_2 is fully determined by K . The other terms have an upper bound independent of the pseudobasis: for t_1 there is Minkowski's bound, while $t_3 \leq 0$ by the arithmetic-geometric inequality, and $t_4 \leq 0$ since the pseudobasis is unital. Sections 4.2 up to 4.5 are aimed at providing an estimate for those terms, rather than just a generic bound.

In Section 4.6, we use our analysis of the four terms in order to conclude with an upper and lower bound on the module-BKZ $\mathbb{Q}$ -slope prediction (Equation (3)), yielding the prediction interval shown in Figure 1.

Remark 6 (Explanation of terminology). Note that $t_3 = 0$ if and only if $|\sigma(\mathbf{s})|^2$ has the same value for all embeddings σ . Thus, t_3 measures how *skewed* these values are. Moreover, t_4 is defined by the norm of ideals $\mathfrak{I} \supseteq \mathcal{O}_K$, implying that $N(\mathfrak{I}) = 1/N(\mathfrak{I}^{-1})$ where $N(\mathfrak{I}^{-1})$ equals the *index* of $\mathfrak{I}^{-1}$ as a subgroup of $\mathcal{O}_K$.

4.1 Module GSA and Corresponding Slope Prediction

Let $\mathcal{M}$ be a rank- r module lattice with unital $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis $((\mathbf{b}_i, \mathfrak{I}_i))_{i=1}^r$ for some $\beta_K \ll r$. Writing $\mathbf{z}_1^*, \dots, \mathbf{z}_{rd}^*$ for the GSO of a corresponding $\mathbb{Z}$ -basis, we recall that the $\mathbb{Q}$ -profile is defined as the sequence $(\ell_i^{\mathbb{Q}})_{i=1}^{rd}$, where:

$$\ell_i^{\mathbb{Q}} := \ln \|\mathbf{z}_i^*\| \quad \forall 1 \leq i \leq rd.$$

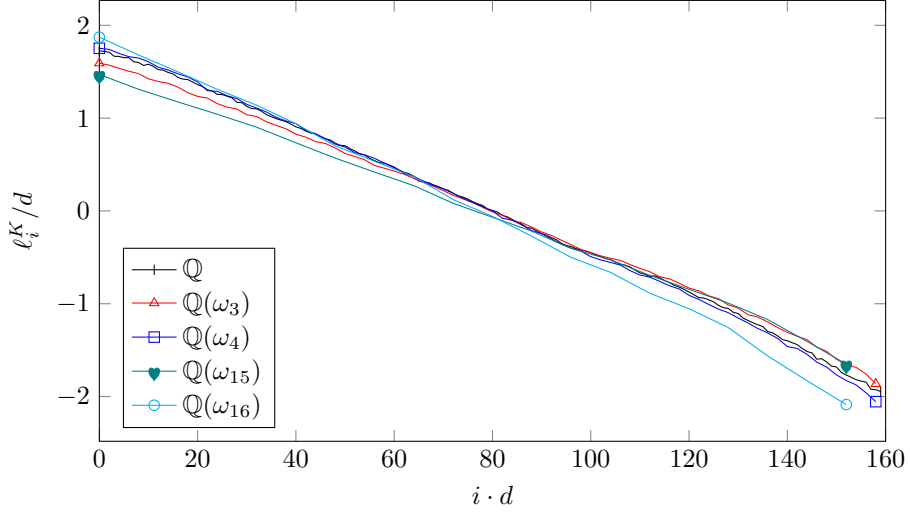
In addition, we consider the K -profile defined as the sequence $(\ell_i^K)_{i=1}^r$, where:

$$\ell_i^K := \ln \det_{\mathbb{Q}}(\mathbf{b}_i^* \cdot \mathfrak{I}_i) = \sum_{j=1}^d \ell_{(i-1)d+j}^{\mathbb{Q}} \quad \forall 1 \leq i \leq r.$$

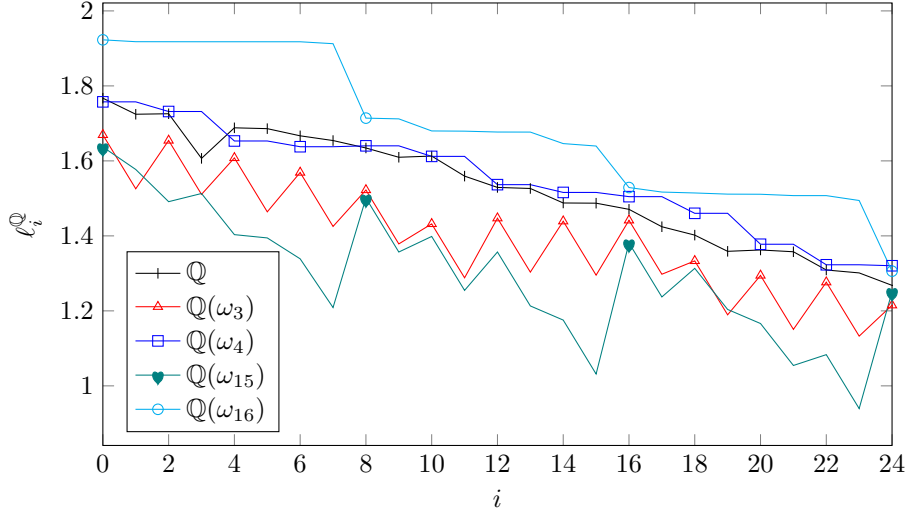
In particular, $\sum_{i=1}^r \ell_i^K = \ln \det_{\mathbb{Q}}(\mathcal{M})$, and each ℓ_i^K/d denotes the average log-norm of the GSO vectors corresponding to the ideal $\mathbf{b}_i \mathfrak{I}_i$.

Figure 4a illustrates the K -profile of mBKZ -reduced bases for the cyclotomic fields $\mathbb{Q}(\omega_3)$, $\mathbb{Q}(\omega_4)$, $\mathbb{Q}(\omega_{15})$, and $\mathbb{Q}(\omega_{16})$. It is compared with $\mathbb{Q}$, the unstructured case. A first observation is that the GSA seems to generalize to the K -profile of mBKZ -reduced bases, up to a tail phenomenon that is well known for regular BKZ [AD21, Def. 9]. However, we notice that the module-BKZ slope varies: it is better (flatter) for $\mathbb{Q}(\omega_3)$ and $\mathbb{Q}(\omega_{15})$, and worse for $\mathbb{Q}(\omega_4)$ and $\mathbb{Q}(\omega_{16})$.¹³

¹³ This discrepancy can be explained by the difference in discriminant gap: cyclotomic fields K with a power-of-two conductor c have $|\Delta_K| = \phi(c)^{\phi(c)}$, whereas a strict inequality holds for conductors $c = 3$ and $c = 15$, contributing to a flatter profile. (See Section 4.3.)



(a) Plot of the normalized K -profile $(\ell_i^K)_{i=1}^r$, averaged over 5 bases.



(b) Plot of $\mathbb{Q}$ -profiles $(\ell_i^{\mathbb{Q}})_{i=1}^{rd}$, zoomed in.

Fig. 4: The profiles resulting from $\text{mBKZ}_K^{\beta_K}$ in dimension $rd = 160$ with blocksize $\beta_K d = 64$ after 30 tours, where $d = \deg(K)$ and $K = \mathbb{Q}(\omega_c)$ for $c = 1, 3, 4, 15, 16$ [by]. The respective degrees are $d = 1, 2, 2, 8, 8$.

In addition, we observe various periodic patterns in the corresponding $\mathbb{Q}$ -profiles shown in Figure 4b. These repeated patterns correspond to the profile of a

(reduced) $\mathbb{Z}$ -basis of $\mathcal{O}_K$, or a small distortion thereof. In the case of $\mathbb{Q}(\omega_4)$, $\mathcal{O}_K \cong \mathbb{Z}^2$ (up to scaling), for which we have $\|\mathbf{b}_1\| = \|\mathbf{b}_2^*\| = 1$: the pattern is perfectly flat on each period. For $\mathbb{Q}(\omega_3)$, $\mathcal{O}_K$ is a scaled hexagonal lattice, with $\|\mathbf{b}_1\| = 1, \|\mathbf{b}_2^*\| = \frac{\sqrt{3}}{2}$. For $\mathbb{Q}(\omega_{16})$ we have $\mathcal{O}_K \cong \mathbb{Z}^8$, but the pattern is not exactly flat, and not perfectly periodic either: $\mathcal{O}_K$ seems to be slightly and randomly *skewed*.

Geometric Series Assumption for Module Lattices. The above observations regarding Figure 4a motivate the following module-lattice analog of the Geometric Series Assumption (GSA). We recall that $\ell_i^K = \ln \det_{\mathbb{Q}}(\mathbf{b}_i^* \cdot \mathcal{I}_i) = \frac{1}{2} \ln |\Delta_K| + \ln N(\mathbf{b}_i^*)N(\mathcal{I}_i)$ for $1 \leq i \leq r$.

Heuristic 3 (Module-Lattice GSA) *Let $\beta_K \ll r$ be sufficiently large. Let $\mathcal{M}$ be a random rank- r module over a number field K , and let $((\mathbf{b}_i, \mathcal{I}_i))_{i=1}^r$ be an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis of $\mathcal{M}$. Then there is a constant $\alpha_K > 1$ (depending on β_K) such that*

$$\mathbb{E}[\ell_i^K] = \mathbb{E}[\ell_1^K] - (i-1) \ln \alpha_K \quad \forall 1 \leq i \leq r. \quad (\text{mGSA})$$

Similar to the unstructured case (Section 3.1), the module invariant $\det_{\mathbb{Q}}(\mathcal{M})$ and the definition of mBKZ reduction allow us to translate Heuristic 3 into a prediction of the *expected K -slope*, defined as $\text{slope}_K(\text{mBKZ}_K^{\beta_K}) := -\ln \alpha_K$.

Remark 7. While the Euclidean GSA (Heuristic 1) may not hold locally, it is globally plausible and allows us to translate the slope of the K -profile back to the slope of the corresponding $\mathbb{Q}$ -profile, namely we predict:

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = \frac{1}{d^2} \text{slope}_K(\text{mBKZ}_K^{\beta_K}).$$

Indeed, we have $\ln \alpha_K = \ell_1^K - \ell_2^K = \sum_{j=1}^d \ell_j^{\mathbb{Q}} - \sum_{j=1}^d \ell_{d+j}^{\mathbb{Q}} = d^2 \ln \alpha_{\mathbb{Q}}$, using Equation (mGSA), the definition of the ℓ_i^K , and Equation (GSA).

Slope Prediction. We now show that the module-lattice GSA (Heuristic 3) yields the following prediction of the expected K -slope, $-\ln \alpha_K$ [47].

Heuristic Claim 2 (K -profile of $\text{mBKZ}_K^{\beta_K}$ under mGSA) *Let K be a number field of degree d . Let $\mathcal{M}$ be a random rank- r module over K , and let $\mathfrak{B}$ be an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis of $\mathcal{M}$ for some sufficiently large $\beta_K \ll r$. Then the module-lattice GSA predicts:*

$$\mathbb{E}[\ell_i^K] = \frac{r+1-2i}{2} \ln \alpha_K + \frac{1}{r} \ln \det_{\mathbb{Q}}(\mathcal{M})$$

for all $1 \leq i \leq r$, where:

$$\ln \alpha_K = \frac{2d}{\beta_K - 1} \left(\mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|] + \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d} \right)$$

$$+ \mathbb{E}_{\mathbf{s}} \left[\ln \frac{\sqrt{d}N(\mathbf{s})^{\frac{1}{d}}}{\|\mathbf{s}\|} \right] + \frac{1}{d} \mathbb{E}_{\mathfrak{I}} [\ln N(\mathfrak{I})] \quad (4)$$

where $\mathbf{s}$ is a shortest vector in one of the random normalized projected module lattices $\mathcal{M}(\mathfrak{B}_{\llbracket j; j+\beta_K-1 \rrbracket})^{(1)}$ for $1 \leq j < r - \beta_K$, and $\mathfrak{I} \supseteq \mathcal{O}_K$ is a fractional ideal in the random pseudobasis $\mathfrak{B}$.

By Remark 7, Heuristic 1 allows us to obtain an expected $\mathbb{Q}$ -slope of

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = -\frac{1}{d^2} \ln \alpha_K,$$

resulting in the slope prediction given in Equation (3).

Justification of Heuristic Claim 2. Let $\mathfrak{B} := ((\mathbf{b}_i, \mathfrak{I}_i))_{i=1}^r$ be the random pseudobasis output by module-BKZ. First, we show that Heuristic 3 implies Equation (4). Let $D_j := \det_{\mathbb{Q}}(\mathcal{M}(\mathfrak{B}_{\llbracket j; j+\beta_K-1 \rrbracket}))$ for some $1 \leq j < r - \beta_K$. By definition of the ℓ_i^K 's and by Equation (mGSA) applied to all $i \in \{1, \dots, \beta_K\}$, we have:

$$\mathbb{E}[\ln(D_j)] = \sum_{i=0}^{\beta_K-1} \mathbb{E}[\ell_{j+i}^K] = \beta_K \left(\mathbb{E}[\ell_j^K] - \frac{\beta_K - 1}{2} \ln \alpha_K \right),$$

so we obtain $\ln \alpha_K = \frac{2}{\beta_K - 1} (\mathbb{E}[\ell_j^K] - \frac{1}{\beta_K} \mathbb{E}[\ln(D_j)])$.

We also have $\ell_j^K = \frac{1}{2} \ln |\Delta_K| + \ln N(\mathbf{b}_j) + \ln N(\mathfrak{I}_j)$, which can be rewritten as:

$$\ell_j^K = d \ln(\|\mathbf{b}_j\|) + \frac{1}{2} \ln \frac{|\Delta_K|}{d^d} + d \ln \frac{\sqrt{d}N(\mathbf{b}_j)^{1/d}}{\|\mathbf{b}_j\|} + \ln N(\mathfrak{I}_j).$$

Hence, $\frac{1}{d}(\ell_j^K - \frac{1}{\beta_K} \ln(D_j)) = \ln \frac{\|\mathbf{b}_j\|}{D_j^{1/\beta_K d}} + \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d} + \ln \frac{\sqrt{d}N(\mathbf{b}_j)^{1/d}}{\|\mathbf{b}_j\|} + \frac{1}{d} \ln N(\mathfrak{I}_j)$.

By definition of mBKZ reduction (Definition 4), $\mathbf{b}_j/D_j^{1/\beta_K d}$ is a shortest vector in the normalized projected module lattice $\mathcal{M}(\mathfrak{B}_{\llbracket j; j+\beta_K-1 \rrbracket})^{(1)}$. In other words, we have shown Equation (4).

To conclude, we apply Equation (mGSA) to all $i \in \{1, \dots, r\}$, giving:

$$\ln \det_{\mathbb{Q}}(\mathcal{M}) = \sum_{i=1}^r \mathbb{E}[\ell_i^K] = r \mathbb{E}[\ell_1^K] - \frac{r(r-1)}{2} \ln \alpha_K.$$

Hence, $\mathbb{E}[\ell_1^K] = \frac{r-1}{2} \ln \alpha_K + \frac{1}{r} \ln \det_{\mathbb{Q}}(\mathcal{M})$, so Equation (mGSA) implies that $\mathbb{E}[\ell_i^K] = \frac{r+1-2i}{2} \ln \alpha_K + \frac{1}{r} \ln \det_{\mathbb{Q}}(\mathcal{M})$, as we wanted to show. $\square$

4.2 Module-Lattice Analog of the Gaussian Heuristic

We approximate the first term $t_1 = \mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]$ in Equation (3) by extending the Gaussian Heuristic to module lattices. This is justified if the first projected

module lattice of an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis behaves like a random rank- β_K module lattice (of the same volume).

Specifically, let K be a number field of degree d . For a positive integer r , we write $\text{gh}_K(r)$ for the expected first minimum of a random rank- r module lattice over K of unit determinant, and denote its logarithm by $\text{lgh}_K(r)$. In the general case, Minkowski's bound allows us to prove $\text{gh}_K(r) \leq 2 \cdot \text{gh}_{\mathbb{Q}}(rd)$ for any number field K . In the case of a cyclotomic field $K = \mathbb{Q}(\omega_c)$ of conductor c and degree $d = \phi(c)$, a recent study [GSVV24, Theorem 38] proved that the first minimum of a (formally well-defined) random rank- r module lattice of unit determinant is asymptotically concentrated around $(\mu_K \cdot \text{vol}(\mathcal{B}_{rd}))^{-rd}$, where we recall that μ_K denotes the number of roots of unity in K and $\mathcal{B}_{rd}$ the rd -dimensional Euclidean unit ball.¹⁴

While this does not predict the exact average, we can attempt to make such a prediction by heuristically ‘merging’ [GSVV24, Theorem 38] and Heuristic 2. Namely, we use the latter as a baseline for $\mathbb{Q}$, and the former to estimate the gap between $\mathbb{Q}$ and K . Because $\mu_{\mathbb{Q}} = 2$, one reaches the following heuristic.

Heuristic 4 ($\lambda_1(\mathcal{M})$ under the Gaussian Heuristic for $\mathcal{M}$ over $\mathbb{Q}(\omega_c)$ [py])
Let $\mathcal{M}$ be a random rank- r module lattice of unit determinant over $K = \mathbb{Q}(\omega_c)$. Its expected first minimum under the Gaussian Heuristic is given by

$$\mathbb{E}[\lambda_1(\mathcal{M})] = \text{gh}_{\mathbb{Q}}(rd) \cdot \left(\frac{\mu_K}{2}\right)^{\frac{1}{rd}}. \quad (\text{mGH})$$

Hence, $\text{lgh}_K(r) = \text{lgh}_{\mathbb{Q}}(rd) + \frac{\ln(\mu_K/2)}{rd}$.

Remark 8. Similar to Heuristic 2, the module-lattice Gaussian Heuristic can also be considered for random module lattices of arbitrary determinant, by appropriate scaling.

This results in the following heuristic estimate:

$$t_1 \approx \text{lgh}_{\mathbb{Q}}(\beta_K d) + \frac{1}{\beta_K d} \ln \frac{\mu_K}{2} = \frac{1}{\beta_K d} \ln \left(\frac{\mu_K}{\text{vol}(\mathcal{B}_{\beta_K d})} \right) + \ln \Gamma \left(1 + \frac{1}{\beta_K d} \right).$$

We verify this prediction experimentally in Figure 5, and note it to be rather accurate even for relatively small rank r , and increasingly accurate as r increases. We note that even for $K = \mathbb{Q}$ the average is slightly underestimated, but the theorem on which we base the module-lattice Gaussian Heuristic only provides a concentration bound, not an expectation. A refined estimate for the average also exists [Che13, BSW18] over $\mathbb{Q}$, and it would be interesting to extend it to other number fields.

¹⁴ We also considered the simpler and specialized Theorem 3 of [GSVV24], but found it to give poor prediction for even conductors c . Recall $\mu_K = 2c$ for odd c and $\mu_K = c$ otherwise.

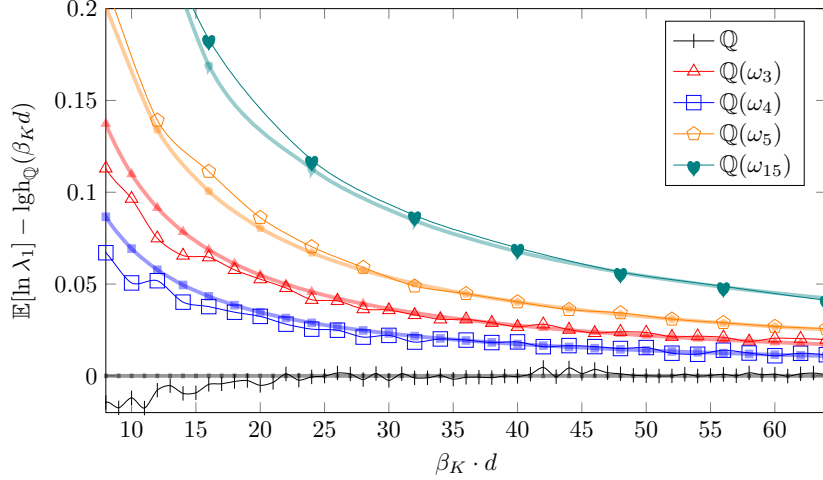


Fig. 5: Logarithmic gap to the Gaussian Heuristic over $\mathbb{Q}$ ($\text{lgh}_{\mathbb{Q}}(rd)$). Thick lines with small marks are predictions ($\text{lgh}_K(r)$), thin lines with large marks are experimental average, taken over 1000 samples $[\text{py}]$.

4.3 Discriminant Gap

The next term in the formula for $\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^\beta)$ is the discriminant gap $t_2 = \frac{1}{2d} \ln \left(\frac{|\Delta_K|}{d^d} \right)$. Ignoring other terms, a smaller discriminant brings the predicted slope closer to 0, thereby contributing to a flatter module-BKZ profile. In the case that K is a cyclotomic field, we have the following explicit formula for its discriminant Δ_K , allowing us to compute the discriminant gap. For $c \in \mathbb{N}$, we write $\mathcal{P}_c$ for the set of distinct prime factors dividing c .

Theorem 9 (Discriminant of Cyclotomic Fields [Was82, Prop. 2.7] $[\text{py}]$). For $c \in \mathbb{N}$, let ω_c be a primitive c -th root of unity. The discriminant Δ_K of $K = \mathbb{Q}(\omega_c)$ is equal to:

$$\Delta_K = (-1)^{\frac{\phi(c)}{2}} c^{\phi(c)} \prod_{p \in \mathcal{P}_c} p^{-\frac{\phi(c)}{p-1}}.$$

where $\mathcal{P}_c$ is the set of prime integers dividing c .

We obtain the following formula for the discriminant gap t_2 of cyclotomic fields, revealing that t_2 merely depends on the set $\mathcal{P}_c$ for the field's conductor c .

Lemma 10 (Discriminant Gap Formula). For $c \in \mathbb{N}$, let ω_c be a primitive c -th root of unity. Then the discriminant gap t_2 of $K = \mathbb{Q}(\omega_c)$ is independent of the exponents in the prime decomposition of c , and equals:

$$t_2 = \frac{1}{2} \sum_{p \in \mathcal{P}_c} \left(\frac{p-2}{p-1} \ln(p) - \ln(p-1) \right).$$

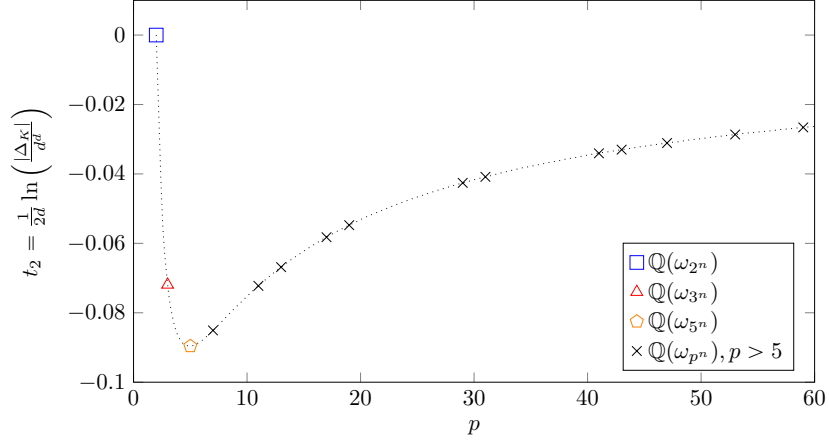


Fig. 6: Discriminant gap t_2 for cyclotomic fields of conductor $c = p^n$, where p is a prime and $n \in \mathbb{N}$.

Proof. For $c \in \mathbb{N}$, let ω_c be a primitive c -th root of unity, and define $K := \mathbb{Q}(\omega_c)$ and $d := \phi(c)$. By Theorem 9, we have $\ln(|\Delta_K|) = d(\ln(c) - \sum_{p \in \mathcal{P}_c} \frac{1}{p-1} \ln(p))$. Therefore, by definition of t_2 , the discriminant gap of K is equal to:

$$t_2 = \frac{1}{2d} \ln(|\Delta_K|) - \frac{1}{2} \ln(d) = \frac{1}{2} \left(\ln(c) - \ln(d) - \sum_{p \in \mathcal{P}_c} \frac{1}{p-1} \ln(p) \right).$$

By definition of ϕ , we have $d = \phi(c) = c \prod_{p \in \mathcal{P}_c} (1 - \frac{1}{p})$, so $\ln(c) - \ln(d) = \sum_{p \in \mathcal{P}_c} (\ln(p) - \ln(p-1))$. The result immediately follows. $\square$

In other words, for cyclotomic fields of conductor c , we have $t_2 = 0$ if c is a power of 2, and $t_2 < 0$ if c has an odd prime factor. In particular, for a composite conductor c , each distinct odd prime factor contributes to decreasing the discriminant gap t_2 , thereby flattening the predicted module-BKZ profile. However, we remark that the quantity $\frac{p-2}{p-1} \ln(p) - \ln(p-1)$ is minimal for $p = 5$ and is increasing for $p \geq 5$. This is illustrated in Figure 6, where we provide the respective values of t_2 for prime conductors $c = 5, 7, 11, 13, 17, 19$, listed in increasing order with respect to t_2 . Asymptotically, for $p \rightarrow \infty$, the contribution of a prime factor p is $\frac{1-\ln(p)}{2p} + O(\frac{1}{p^2})$.

We also numerically checked in **sage** whether subfields of cyclotomic fields could give good values of t_2 up to conductor 105_[.py]. They turned out to always be equal or larger than for the original cyclotomic.

4.4 Skewness Gap

Next, we consider the skewness gap $t_3 = \mathbb{E} \left[\ln \left(\frac{\sqrt{d} N(\mathbf{s})^{1/d}}{\|\mathbf{s}\|} \right) \right]$, where $\mathbf{s}$ is a shortest vector in the random lattices encountered by module-BKZ. We recall that

t_3 measures how *skewed* the values $|\sigma(\mathbf{s})|^2$ are for the embeddings σ . By construction, this quantity is invariant by scaling the value $\|\mathbf{s}\|$, so we only need to model its direction. Quite naturally, we are tempted to model the direction as being uniform over the $(\beta_K - 1)$ -dimensional unit sphere $\mathcal{S}(K_{\mathbb{R}}^{\beta_K})$, forgetting about the fact that $\mathbf{s}$ should be a shortest vector of a module (which certainly constraints its direction [CDPR16]). Conveniently, we model $\mathbf{s} \in K_{\mathbb{R}}^{\beta_K}$ itself as a spherical Gaussian, and we denote this modeled skewness gap by t_3 . This makes our analysis similar to that of [DvW21, Lemma 4.4], generalized to arbitrary number fields and module ranks.

Let us first express the independent real Gaussian variables more explicitly using the definition of the Euclidean norm. For $\mathbf{s} = (s_1, \dots, s_{\beta_K})$, we have:

$$\|\mathbf{s}\|^2 = \text{Tr}(\langle \mathbf{s}, \mathbf{s} \rangle_K) = \sum_{\sigma \in \mathcal{E}} \sum_{j=1}^{\beta_K} \sigma(s_j) \sigma(\overline{s_j}).$$

For each $j \in [1; \beta_K]$, $\sigma_{\mathbb{R}} \in \mathcal{E}_{\mathbb{R}}$, and $\sigma_{\mathbb{C}} \in \mathcal{E}_{\mathbb{C}}^+$, we define $A_{\sigma_{\mathbb{R}}, j} := \sigma_{\mathbb{R}}(s_j)$, $B_{\sigma_{\mathbb{C}}, j} := \sqrt{2} \cdot \Re(\sigma_{\mathbb{C}}(s_j))$, and $C_{\sigma_{\mathbb{C}}, j} := \sqrt{2} \cdot \Im(\sigma_{\mathbb{C}}(s_j))$. Then:

$$\|\mathbf{s}\|^2 = \sum_{j=1}^{\beta_K} \left(\sum_{\sigma \in \mathcal{E}_{\mathbb{R}}} A_{\sigma, j}^2 + \sum_{\sigma \in \mathcal{E}_{\mathbb{C}}^+} (B_{\sigma, j}^2 + C_{\sigma, j}^2) \right). \quad (5)$$

On the other hand, we have:

$$N(\mathbf{s})^2 = \prod_{\sigma \in \mathcal{E}} \sum_{j=1}^{\beta_K} \sigma(s_j) \sigma(\overline{s_j}) = \left(\prod_{\sigma \in \mathcal{E}_{\mathbb{R}}} \sum_{j=1}^{\beta_K} A_{\sigma, j}^2 \right) \left(\prod_{\sigma \in \mathcal{E}_{\mathbb{C}}^+} \frac{1}{2} \sum_{j=1}^{\beta_K} (B_{\sigma, j}^2 + C_{\sigma, j}^2) \right)^2.$$

It is therefore $A_{\sigma, j}$, $B_{\sigma, j}$, and $C_{\sigma, j}$ in Equation (5) that we model as independent Gaussian variables, say of variance 1. The terms $\sum_{j=1}^{\beta_K} A_{\sigma, j}^2$, $\sum_{j=1}^{\beta_K} B_{\sigma, j}^2$, and $\sum_{j=1}^{\beta_K} C_{\sigma, j}^2$ in the above formulas then follow a χ^2 distribution, and so does $\|\mathbf{s}\|^2$. The expected logarithm of a χ^2 distribution relates to the digamma function, denoted ψ , via $\mathbb{E}[\ln \chi_a^2] = \psi(a/2) + \ln 2$, where a is the number of degrees of freedom. We proceed with the calculation [27], using $d = d_{\mathbb{R}} + 2d_{\mathbb{C}}$ to cancel some $\ln 2$ terms:

$$\begin{aligned} \tilde{t}_3 &:= \mathbb{E}_{\mathbf{s} \sim \mathcal{S}(K_{\mathbb{R}}^{\beta_K})} \left[\ln \left(\frac{\sqrt{d} N(\mathbf{s})^{\frac{1}{d}}}{\|\mathbf{s}\|} \right) \right] \\ &= \frac{\ln d}{2} + \frac{d_{\mathbb{R}} \mathbb{E}[\ln \chi_{\beta_K}^2] + 2d_{\mathbb{C}} (\mathbb{E}[\ln \chi_{2\beta_K}^2] - \ln 2)}{2d} - \frac{\mathbb{E}[\ln \chi_{\beta_K d}^2]}{2} \\ &= \frac{\ln d}{2} + \frac{d_{\mathbb{R}} \psi(\beta_K/2) + 2d_{\mathbb{C}} (\psi(\beta_K) - \ln 2)}{2d} - \frac{\psi(\beta_K d/2)}{2}. \end{aligned} \quad (6)$$

When K is a totally imaginary field, such as a cyclotomic field of conductor $c > 2$, we have $d_{\mathbb{R}} = 0$ and $2d_{\mathbb{C}} = d$. In fact, Equation (6) equals 0 when K is an

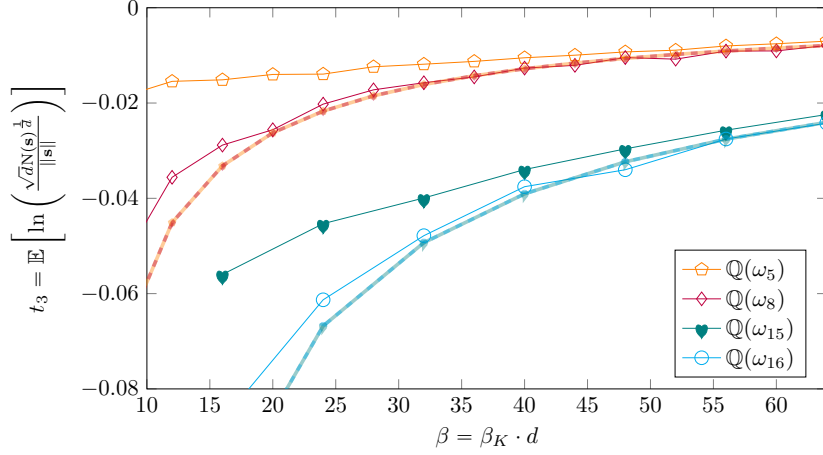


Fig. 7: Skewness gap (term t_3). Thick translucent lines with small marks correspond to the prediction $\tilde{t}_3$ of our model in Equation (6), thin lines with large marks are the experimental average, taken over 1000 samples [·_{pf}]. We remark that as the prediction depends only on $d_{\mathbb{R}}$ and $d_{\mathbb{C}}$, predictions for different rings can overlap.

imaginary *quadratic* field ($d = 2$). Moreover, for $K = \mathbb{Q}$, our model adequately predicts no skewness as well.

Figure 7 suggests that the model and experiments converge for large values of β_K , but $\tilde{t}_3$ significantly underestimates the skewness term t_3 for smaller β_K . This means that our model is not exactly accurate. In particular, it does not account for the fact that $\mathbf{s}$ must be a shortest vector of a module lattice, and therefore of $\mathbf{s}\mathcal{O}_K$.

Let us fix $\|\mathbf{s}\| = 1$ by rescaling. In our model, $N(\mathbf{s})^{1/d}$, has a small probability of being arbitrarily small, for example if the $A_{i,\sigma}$ are small enough for a fixed σ and all i (think of points on a sphere of dimension $\beta_K d$ close to a hyperplane of dimension β_K). However, such a situation is forbidden by Minkowski's bound: if $N(\mathbf{s})^{1/d}$ gets arbitrarily small, $\det_{\mathbb{Q}}(\mathbf{s}\mathcal{O}_K)$ does as well, putting an arbitrary small upper bound on its minimal distance.

4.5 Index Gap

Finally, we consider the index gap $t_4 = \frac{1}{d} \mathbb{E}[\ln(N(\mathcal{J}))]$. By construction, the pseudobasis is unital, so $\mathcal{O}_K \subseteq I$ and thus $\ln(N(\mathcal{J})) \leq 0$.

Let us first clarify that $\ln(N(\mathcal{J})) < 0$ does not require $\mathcal{O}_K$ to have non-principal ideals: the shortest vector of a principal ideal may indeed not be a generator. There are even cases where the shortest generator is subexponentially larger than the the shortest vector [CDPR16, Lemma 6.2].

We now provide a lower bound $\tilde{t}_4$ on t_4 using a density-based argument based on the Dedekind zeta function ζ_K , in a similar way as [ABD16, Sec 2.2], [DPPvW22, App. A], and [DvW21, Claim 4.2]. We assume here that the random rank- β_K module admits a K -basis $\mathbf{B}$ rather than a pseudobasis, and let $\mathbf{v} = \mathbf{B}\mathbf{x}$ be a shortest vector. The ideal $\mathfrak{J}$ constructed in the module-BKZ algorithm is then equal to $\gcd(\{x_i \mathcal{O}_K\}_i)^{-1}$.

Following standard analysis, and assuming that the x_i 's are random large elements of $\mathcal{O}_K$, the probability that this gcd is a multiple of an ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ is $N(\mathfrak{a})^{-\beta_K}$. Writing $\mathcal{D}$ for the distribution of the $\mathfrak{J}$ under our model, and decomposing $\mathfrak{J}$ over the prime ideals of $\mathcal{O}_K$, we obtain [py]:

$$\begin{aligned}
\tilde{t}_4 &:= \frac{1}{d} \mathbb{E}_{\mathfrak{J} \sim \mathcal{D}} [\ln N(\mathfrak{J})] \\
&= -\frac{1}{d} \sum_{\mathfrak{p}} \sum_{i \in \mathbb{N}} i \cdot (\Pr[\mathfrak{J}^{-1} \subset \mathfrak{p}^i] - \Pr[\mathfrak{J}^{-1} \subset \mathfrak{p}^{i+1}]) \cdot \ln N(\mathfrak{p}) \\
&= -\frac{1}{d} \sum_{\mathfrak{p}} \sum_{i \in \mathbb{N}} i \cdot (N(\mathfrak{p})^{-i\beta_K} - N(\mathfrak{p})^{-(i+1)\beta_K}) \cdot \ln N(\mathfrak{p}) \\
&= -\frac{1}{d} \sum_{\mathfrak{p}} \sum_{i \in \mathbb{N}} N(\mathfrak{p})^{-i\beta_K} \ln N(\mathfrak{p}) \\
&= -\frac{1}{d} \sum_{\mathfrak{p}} \frac{\ln N(\mathfrak{p})}{N(\mathfrak{p})^{\beta_K} - 1} = \frac{1}{d} \frac{\zeta'_K(\beta_K)}{\zeta_K(\beta_K)} \tag{7}
\end{aligned}$$

where ζ_K denotes the Dedekind zeta function of K , and ζ'_K/ζ_K is its logarithmic derivative [py]. Here, we used a telescoping identity $\sum_{i=1}^{\infty} i \cdot (x^i - x^{i+1}) = \sum_{i=1}^{\infty} x^i$.

However, over $\mathbb{Q}$, $\mathbb{Q}(\omega_3)$, and $\mathbb{Q}(\omega_4)$, the algebraic norm is an increasing function of the Euclidean norm, so shortest vectors and generators of rank-1 ideals always coincide: it can never be that $N(\mathfrak{J}) < 1$ despite the above analysis saying it happens with nonzero probability. The previous analysis fails to capture that $\mathbf{v}$ is short, which lowers the probability of finding divisors in this gcd.

The failure of our modeled index gap $\tilde{t}_4$ is blatant in Figure 8: for small rank β_K it predicts a significant index gap, while for conductor $c \in \{1, 3, 4, 5, 8, 15\}$ the index gap was always trivial over 1000 samples. We did encounter $N(\mathfrak{J}) < 1$ for cyclotomic conductor 16, however, but still much less often than predicted. Overall, it seems better to treat this model as a lower bound on the index gap.

Yet, as we discuss in Section 5.1, the modeled index gap converges exponentially fast to 0 as a function of β_K anyway, so this term is actually rather well controlled for large blocksizes.

4.6 Putting It All Together

We bring our analysis of the terms t_1, t_2, t_3, t_4 together and conclude on our prediction bounds for the module-BKZ $\mathbb{Q}$ -slope, as illustrated in Figure 1.

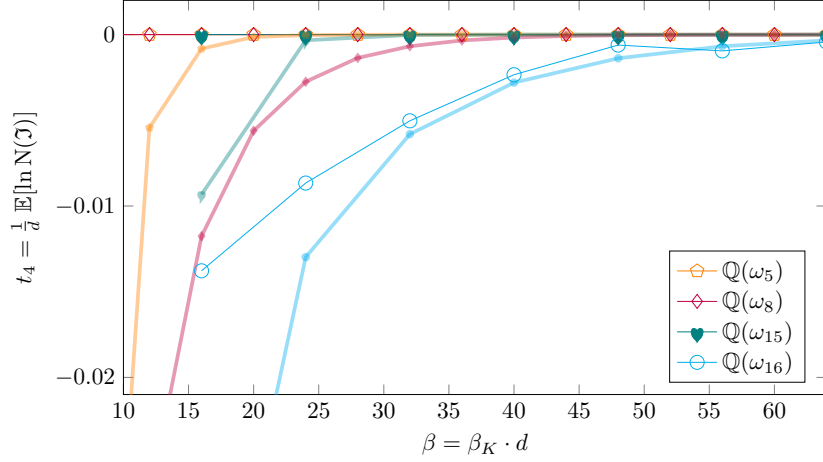


Fig. 8: Index gap (term t_4). Thick translucent lines with small marks correspond to the prediction $\hat{t}_4$ of our model in Equation (7), thin lines with large marks are the experimental average, taken over 1000 samples $[\cdot]_{\text{py}}$.

Let $K = \mathbb{Q}(\omega_c)$ be a cyclotomic field of conductor $c > 2$ and degree $d = \phi(c)$. We have shown that the module-BKZ $\mathbb{Q}$ -slope satisfies

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = -\frac{2}{\beta_K d - d}(t_1 + t_2 + t_3 + t_4)$$

where $t_1 = \text{lgh}_{\mathbb{Q}}(\beta_K d) + \frac{\ln c}{\beta_K d}$, $t_2 = \frac{1}{2d} \ln\left(\frac{|\Delta_K|}{d^d}\right)$, $\frac{\ln(d/2) + \psi(r) - \psi(rd/2)}{2} \leq t_3 \leq 0$, and $\frac{1}{d} \frac{\zeta'_K(\beta_K)}{\zeta_K(\beta_K)} \leq t_4 \leq 0$. We rely on the module-lattice Gaussian Heuristic for t_1 , while the lower bounds for t_3 and t_4 are based on the spherical model and the density-based argument (resp.). In contrast, the formula for t_2 does not rely on any heuristic and can be written explicitly in terms of c 's distinct prime factors.

We implemented these formulas to predict the module-BKZ $\mathbb{Q}$ -slope interval and compared it against the results from our module-BKZ implementation. This prediction interval and its comparison with practice are shown in Figure 1 in Section 1 for some cyclotomic fields. Looking at the plots, our prediction interval seems accurate for a rather large range of block sizes, showing a significant gain for $\mathbb{Q}(\omega_3)$ and $\mathbb{Q}(\omega_{15})$, and a significant loss for $\mathbb{Q}(\omega_8)$, compared to $\mathbb{Q}$. We observe a minor quantitative misfit for $\mathbb{Q}$ and $\mathbb{Q}(\omega_3)$, which might be due to head and tail phenomena that our module-lattice GSA does not account for, but at least qualitatively the gain for $\mathbb{Q}(\omega_3)$ is experimentally confirmed. Such a misfit also appears in unstructured BKZ [YD17,BSW18], and can possibly be overcome by a tail-adapted refinement and simulation (recall Open Questions 1, 2).

5 Asymptotic Analysis of the Blocksize Gain

Let $\mathcal{M}$ be a random rank- r module lattice over a degree- d number field K . In this section, we compare the predicted BKZ and module-BKZ slopes by determining the blocksize β_{eq} for which $\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta}) = \text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_{\text{eq}}/d})$, and analyse how it behaves as $\beta \rightarrow \infty$. Concretely, this allows us to quantify the gain or loss of module-BKZ as the difference $\beta_{\text{eq}} - \beta$, as illustrated in Figure 2 in Section 1.

We recall that given an SVP oracle for (unstructured) lattices of full rank β , the predicted BKZ slope and mBKZ slope are respectively:

$$\begin{aligned}\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta}) &= -\frac{2}{\beta-1} \text{lgh}_{\mathbb{Q}}(\beta), \\ \text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta/d}) &= -\frac{2}{\beta-d} (t_1 + t_2 + t_3 + t_4),\end{aligned}$$

where the terms t_1, t_2, t_3, t_4 are defined with respect to the K -rank $\beta_K := \beta/d$.

5.1 Asymptotic Behavior of Each Term

First, we detail the asymptotic contribution of each t_i in order to identify the leading asymptotic terms.

Module-Lattice Gaussian Heuristic t_1 . To analyze the term $t_1 \approx \text{lgh}_{\mathbb{Q}}(\beta) + \frac{1}{\beta} \ln \frac{\mu_K}{2}$ (under Heuristic 4), we recall the expression of the Gaussian heuristic $\text{gh}_{\mathbb{Q}}(\beta) = \Gamma(1 + \frac{1}{\beta})(\frac{2}{\text{vol } \mathcal{B}_{\beta}})^{\frac{1}{\beta}}$ where Γ denotes the Gamma function and $\mathcal{B}_{\beta}$ the β -dimensional Euclidean unit ball. By the Taylor series expansion for $z > 0$, $\ln \Gamma(1+z) = -\gamma z + o(z)$ as $z \rightarrow 0$, where γ denotes the Euler-Mascheroni constant. Moreover, we have $\ln \Gamma(z) = z \ln(z) - z - \frac{\ln z}{2} + o(\ln z)$ as $z \rightarrow +\infty$. Since $\text{lgh}_{\mathbb{Q}}(\beta) = \ln \Gamma(1 + \frac{1}{\beta}) + \frac{1}{\beta} \ln \Gamma(1 + \frac{\beta}{2}) + \ln \frac{2^{\frac{1}{\beta}}}{\sqrt{\pi}}$, we obtain:

$$\text{lgh}_{\mathbb{Q}}(\beta) = \frac{\ln(\beta+2)}{2} - \frac{\ln(2\pi e)}{2} + \frac{\ln \beta}{2\beta} + o\left(\frac{\ln \beta}{\beta}\right).$$

Discriminant Gap t_2 . The second term, $t_2 = \frac{1}{2d} \ln \left(\frac{|\Delta_K|}{d^d} \right)$, depends only on K , and is therefore constant in β .

Skewness Gap t_3 . Following our analysis in Section 4.4, we assume that t_3 converges to $\tilde{t}_3$ as $\beta \rightarrow \infty$, where $\tilde{t}_3 = \frac{\ln d}{2} + \frac{d_{\mathbb{R}} \cdot \psi(\beta/2d) + 2d_{\mathbb{C}} \cdot (\psi(\beta/d) - \ln 2)}{2d} - \frac{\psi(\beta/2)}{2}$ for the digamma function ψ . Since the digamma function behaves asymptotically as $\psi(z) = \ln(z) - \frac{1}{2z} + o(\frac{1}{z})$, we obtain:

$$t_3 = \frac{1 - d_{\mathbb{R}} - d_{\mathbb{C}}}{2\beta} + o\left(\frac{1}{\beta}\right) = o\left(\frac{\ln \beta}{\beta}\right).$$

Index Gap t_4 . Following our analysis in Section 4.5, we assume that t_4 converges to $\tilde{t}_4$ as $\beta \rightarrow \infty$, where $\tilde{t}_4 = -\frac{1}{d} \sum_{\mathfrak{p}} \frac{\ln N(\mathfrak{p})}{N(\mathfrak{p})^{\beta_K-1}}$ with $\mathfrak{p}$ ranging over the prime ideals of $\mathcal{O}_K$ and $\beta_K := \beta/d$. Combining Lemma 11 and Lemma 12 in Appendix A.1 then yields $t_4 \geq -\sum_p \frac{\ln(p)}{p^{\beta_K-1}}$, where p ranges over the prime integers $p \in \mathbb{N}$ such that $\mathfrak{p} \mid p\mathcal{O}_K$ for some prime ideal $\mathfrak{p} \subset \mathcal{O}_K$. Thus, writing $\mathcal{P}$ for the set of prime integers in $\mathbb{N}$, we obtain:

$$t_4 \geq -\sum_{p \in \mathcal{P}} \frac{\ln(p)}{p^{\beta_K-1}} \geq -\sum_{p \in \mathcal{P}} \frac{2 \ln(p)}{p^{\beta_K}} \geq -\sum_{p \in \mathcal{P}} \frac{1}{p^{\beta_K-1}} \geq -\frac{1}{2^{\beta_K-1}} - \int_2^\infty \frac{du}{u^{\beta_K-1}}.$$

In particular, for $\beta_K \geq 3$, we have $-\frac{1}{2^{\beta_K-1}} - \frac{1}{(\beta_K-2)2^{\beta_K-2}} \leq t_4 \leq 0$, from which we conclude that

$$t_4 = -\frac{1}{2^{\beta_K-1}} + o\left(\frac{1}{2^{\beta_K}}\right) = o\left(\frac{1}{\beta}\right).$$

5.2 Asymptotic Gain Compared to Unstructured BKZ

Finally, we use our asymptotic analysis to show the asymptotic relation between β and β_{eq} , when β_{eq} is chosen such that $\text{slope}_{\mathbb{Q}}(\text{BKZ}^\beta) = \text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_{eq}/d})$.

Heuristic Claim 3 *Let K be a field of degree d , and $\beta_{eq} = \beta_{eq}(\beta)$ be such that $\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_{eq}/d}) = \text{slope}_{\mathbb{Q}}(\text{BKZ}^\beta)$. Then, as $\beta \rightarrow +\infty$, our heuristic analysis predicts:*

$$\beta_{eq} = \beta + \ln\left(\frac{|\Delta_K|}{d^d}\right) \frac{\beta}{d \ln \frac{\beta}{2\pi e}} + d - 1 + o(1). \quad (8)$$

A justification can be found in Appendix A.2.

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A Appendix

A.1 Lemmas for the Decomposition of Prime Integers over Number Fields

The following facts are fairly standard, and are presented in different form in [Neu92, Sam13].

Lemma 11. *Let $p \in \mathbb{N}$ be a prime integer, and K a field of degree d . Then, $p\mathcal{O}_K$ may be decomposed in a product of at most d prime ideals $\mathfrak{p}$ such that $N(\mathfrak{p}) = p^k$ for some integer k in $\llbracket 1; d \rrbracket$.*

Proof. Let $p\mathcal{O}_K = \prod_{i \in \mathcal{I}} \mathfrak{p}_i^{\nu_i}$ be the unique decomposition of $p\mathcal{O}_K$ in a product of prime ideals of $\mathcal{O}_K$. Then: $N(p\mathcal{O}_K) = N(\prod_{i \in \mathcal{I}} \mathfrak{p}_i^{\nu_i}) = \prod_{i \in \mathcal{I}} N(\mathfrak{p}_i)^{\nu_i}$, and as $N(p\mathcal{O}_K) = p^d$, one has: $\prod_{i \in \mathcal{I}} N(\mathfrak{p}_i)^{\nu_i} = p^d$. Hence, for each $i \in \mathcal{I}$, there exists $k_i \in \llbracket 1; d \rrbracket$ such that $N(\mathfrak{p}_i) = p^{k_i}$, and $\sum_{i \in \mathcal{I}} k_i \nu_i = d$.

Lemma 12. *For every prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, there exists a unique prime integer $p \in \mathbb{N}$ such that $\mathfrak{p} | p\mathcal{O}_K$, and moreover $N(\mathfrak{p}) = p^k$ for some $k \in \llbracket 1; d \rrbracket$.*

Proof. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$. Then, $\mathfrak{p} \cap \mathbb{Z}$ is a prime ideal $\langle p \rangle$ in $\mathbb{Z}$ (cf. proof of Theorem 3.1 in [Neu92]). Hence $p\mathcal{O}_K \subset \mathfrak{p}$, and $\mathfrak{p} | p\mathcal{O}_K$. Then, Lemma 11 states that $N(\mathfrak{p}) = p^k$ for some $k \in \llbracket 1; d \rrbracket$, and there may be no other prime $q \in \mathbb{N}$, $q \neq p$ such that $\mathfrak{p} | q\mathcal{O}_K$, as then one would have $N(\mathfrak{p}) = q^\kappa$, $\kappa \in \llbracket 1; d \rrbracket$, which would contradict $N(\mathfrak{p}) = p^k$.

A.2 Justification of Heuristic Claim 3

Here, we show how Heuristic Claim 3 is derived from the GSA, the Gaussian heuristic, and our modeling of the distribution of random lattices.

Justification. We first handle the heaviest asymptotic term in the general case. By definition of β_{eq} and by the asymptotic results from Section 5.1, we have:

$$\frac{\text{lgh}_{\mathbb{Q}}(\beta)}{\beta - 1} = \frac{t_1 + t_2 + t_3 + t_4}{\beta_{\text{eq}} - d}$$

where we remark that the t_i are functions of β_{eq} , with $t_1 = \text{lgh}_K\left(\frac{\beta_{\text{eq}}}{d}\right) = \text{lgh}_{\mathbb{Q}}(\beta_{\text{eq}}) + o\left(\frac{\ln(\beta)}{\beta}\right)$, $t_2 = \frac{1}{2d} \ln\left(\frac{|\Delta_K|}{d^d}\right)$ is constant, and our asymptotic analysis from Section 5.1 shows that t_3 and t_4 are both $o\left(\frac{\ln \beta}{\beta}\right)$ as β grows. As $2 \text{lgh}_{\mathbb{Q}}(\beta) = \ln(\beta + 2) - \ln(2\pi e) + \frac{\ln \beta}{\beta} + o\left(\frac{\ln \beta}{\beta}\right) = \ln(\beta) - \ln(2\pi e) + \frac{3 \ln \beta}{\beta} + o\left(\frac{\ln \beta}{\beta}\right)$ when $\beta \rightarrow +\infty$, we obtain:

$$\beta_{\text{eq}} = d + (\beta - 1) \frac{2 \text{lgh}_{\mathbb{Q}}(\beta_{\text{eq}}) + 2t_2 + o\left(\frac{\ln \beta}{\beta}\right)}{2 \text{lgh}_{\mathbb{Q}}(\beta)}$$

$$= d + \beta \cdot \frac{\ln(\beta_{\text{eq}}) - \ln(2\pi e) + 2t_2 + \frac{3\ln\beta}{\beta} + o\left(\frac{\ln\beta}{\beta}\right)}{\ln(\beta) - \ln(2\pi e) + \frac{3\ln\beta}{\beta} + o\left(\frac{\ln\beta}{\beta}\right)} - 1 \cdot \frac{\ln(\beta_{\text{eq}}) + o(\ln\beta)}{\ln(\beta) + o(\ln(\beta))}.$$

Denoting the numerator of the second term by $\alpha(\beta_{\text{eq}}) := \ln(\beta_{\text{eq}}) - \ln(2\pi e) + 2t_2 + \frac{3\ln\beta}{\beta} + o\left(\frac{\ln\beta}{\beta}\right)$, a series expansion of the denominators (expanding $\frac{1}{1+x}$ for $x \rightarrow 0$, and using $\beta_{\text{eq}} = \beta + o(\beta)$) gives:

$$\beta_{\text{eq}} = \beta \cdot \frac{\alpha(\beta_{\text{eq}})}{\ln \frac{\beta}{2\pi e}} \left(1 - \frac{3\ln\beta}{\beta(\ln(\beta) - \ln(2\pi e))} + o\left(\frac{1}{\beta}\right) \right) + d - 1 + o(1)$$

As $-\frac{\ln(2\pi e)}{\beta(\ln(\beta) - \ln(2\pi e))} = o\left(\frac{1}{\beta}\right)$, we obtain:

$$\begin{aligned} \beta_{\text{eq}} &= \beta \cdot \frac{\alpha(\beta_{\text{eq}})}{\ln \frac{\beta}{2\pi e}} \left(1 - \frac{3(\ln\beta - \ln(2\pi e))}{\beta(\ln(\beta) - \ln(2\pi e))} + o\left(\frac{1}{\beta}\right) \right) + d - 1 + o(1) \\ &= \beta \cdot \frac{\alpha(\beta_{\text{eq}})}{\ln \frac{\beta}{2\pi e}} \left(1 - \frac{3}{\beta} + o\left(\frac{1}{\beta}\right) \right) + d - 1 + o(1) \end{aligned} \quad (9)$$

where:

$$\frac{\alpha(\beta_{\text{eq}})}{\ln \frac{\beta}{2\pi e}} = \frac{\ln(\beta_{\text{eq}}) - \ln(2\pi e) + 2t_2 + \frac{3\ln\beta}{\beta}}{\ln(\beta) - \ln(2\pi e)} + o\left(\frac{1}{\beta}\right).$$

As $\ln(\beta_{\text{eq}}) = \ln\left(\beta \left(\frac{\ln(\beta_{\text{eq}})}{\ln\beta} + \frac{2t_2}{\ln\beta} + o\left(\frac{1}{\ln\beta}\right)\right)\right)$, recursive composition with the contractive logarithm function yields:

$$\frac{\alpha(\beta_{\text{eq}})}{\ln \frac{\beta}{2\pi e}} = 1 + \frac{2t_2 + \frac{3\ln\beta}{\beta}}{\ln(\beta) - \ln(2\pi e)} + o\left(\frac{1}{\beta}\right) = 1 + \frac{2t_2}{\ln \frac{\beta}{2\pi e}} + \frac{3}{\beta} + o\left(\frac{1}{\beta}\right).$$

Plugging this into Equation (9) then yields:

$$\beta_{\text{eq}} = \beta \left(1 + \frac{2t_2}{\ln \frac{\beta}{2\pi e}} + \frac{3}{\beta} + o\left(\frac{1}{\beta}\right) \right) \left(1 - \frac{3}{\beta} + o\left(\frac{1}{\beta}\right) \right) + d - 1 + o(1).$$

Finally, expanding provides:

$$\begin{aligned} \beta_{\text{eq}} &= \beta \left(1 + \frac{2t_2}{\ln \frac{\beta}{2\pi e}} + o\left(\frac{1}{\beta}\right) \right) + d - 1 + o(1) \\ &= \beta + 2t_2 \cdot \frac{\beta}{\ln \frac{\beta}{2\pi e}} + d - 1 + o(1). \end{aligned}$$

□